Novel Leukemia Detection

A block-by-block explainer of Figure 1, from the blood-smear database to the final detection and the GUI

THE PROJECT IN ONE PARAGRAPH

A photograph of a blood smear goes in. The software first cleans the picture and makes its edges clear (**HDS**), then finds and outlines the white blood cells with a curvature-based structure detector and a colour-based segmenter (**APC**), then breaks the picture into layers of fine-to-coarse detail and keeps the meaningful layers (**BEEMD pure-IMF selection**). A neural-network classifier (**Classification**) studies the cells and the final block (**Cancer Detection**) reports whether the sample looks benign or is one of three stages of acute lymphoblastic leukemia (ALL). A separately built **GUI** lets a user load an image and read the result.

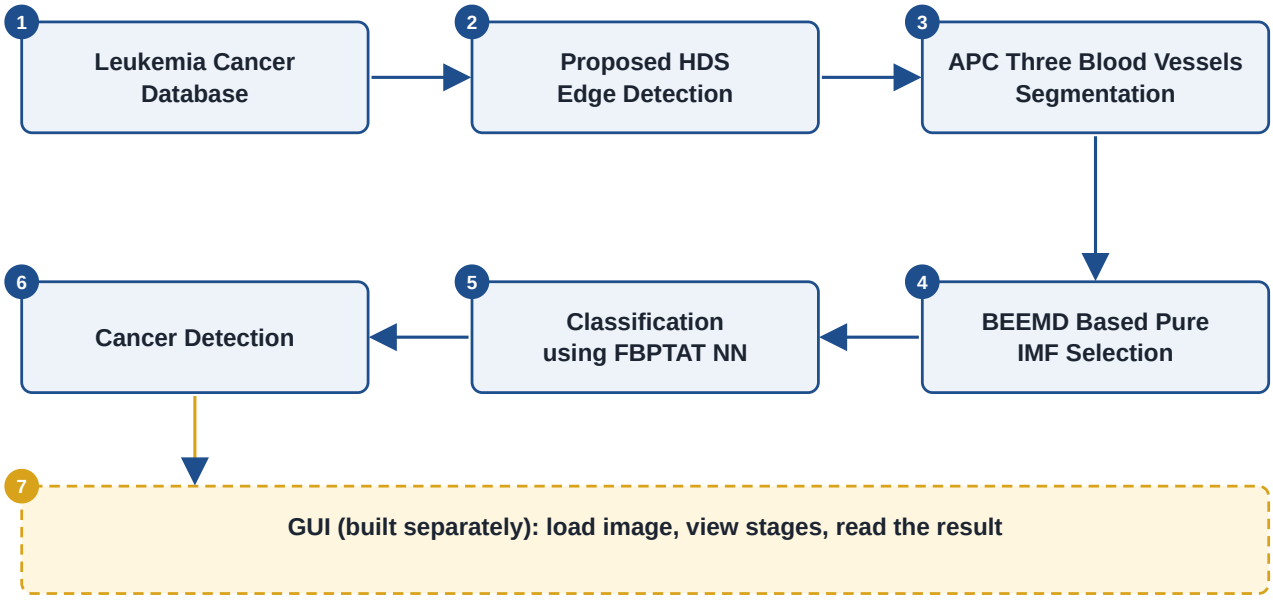


Figure 1. Novel Leukemia Detection (the block diagram as submitted), with the separately built GUI shown as block 7. Every section below follows these numbered blocks and uses the block names exactly as drawn.

What is proposed and what was built and measured

Figure 1 is the proposed method. The accompanying software and paper implement and measure it, with a few deliberate differences. This table is the key to reading everything that follows: the left column is the idea behind each block; the right column is what the working system does and what was measured.

Block	Idea in the proposal	What was built and measured here
1 Database	C-NMC 2019 / ISBI ALL challenge images (10,661 training cell images, 73 subjects; two test sets).	A public four-class peripheral blood smear dataset: **3,256** images of 224×224 pixels (Benign 504, Early 985, Pre 963, Pro 804).

2 HDS	Hybrid Diffusion-Steered denoising and edge detection (Total Variation + Perona-Malik).	Implemented in a **stabilised** form, benchmarked on noise, and run on all 3,256 images.
3 APC	Adaptive Principal Curvature segmentation, scored by Dice / Jaccard against manual masks.	APC response implemented as specified. Cell outlines come from a colour-based segmenter. The dataset has no manual masks, so Dice / Jaccard were not computed.
4 BEEMD	Decompose into intrinsic mode functions (IMFs) and keep the pure ones.	BEEMD with fast order-statistic envelopes and a scale/energy rule for pure IMFs, run on all images.
5 Classification	FBPTAT: a recurrent network trained with fractional-order back-propagation.	A **frozen DinoBloom-S encoder with an attention head** was trained and measured. The FBPTAT network itself was not implemented, so **no FBPTAT numbers are reported**.
6 Cancer Detection	Accuracy, sensitivity, specificity on the database, then on real patient samples.	Measured with a leakage-aware five-fold protocol on the database. Testing on real patient samples was not done.
7 GUI	User interface.	Built separately; described here only through how it connects to the system.

The proposal plans MATLAB as the tool; the system described here is written in Python. All numbers in this document come from runs of that Python system.

Block 1: Leukemia Cancer Database

> **IN ONE LINE**
> The collection of labelled blood-smear photographs that everything else learns from and is tested on.

Why leukemia, and why blood smears

Leukemia is a cancer of the blood-forming cells in the bone marrow and blood. It has four main types (ALL, CLL, AML, CML). This work targets **acute lymphoblastic leukemia (ALL)**, a cancer of the lymphocyte line that makes up roughly a quarter of childhood cancers. Symptoms appear late, so the disease is often found at an advanced stage, when survival is lower. A **peripheral blood smear** (a drop of blood spread on a glass slide, stained and photographed under a microscope) is cheap and quick, which makes it a natural place to build a screening aid. It is only a screening step: confirming and subtyping leukemia needs bone-marrow examination and laboratory tests.

The database in this project

Class	Meaning	Images	Cells found
Benign	Hematogones: normal immature lymphocyte precursors that look like leukemia cells	504	mean 7.9 per image
Early	Early pre-B lymphoblasts (malignant)	985	mean 7.5 per image
Pre	Pre-B lymphoblasts (malignant)	963	mean 20.5 per image
Pro	Pro-B lymphoblasts (malignant)	804	mean 6.2 per image
Total		**3,256**	**36,167 cells**

Each image is 224×224 pixels, with a median of 8 cells and at most 41. The class labels come from the dataset documentation, which states that experts labelled the images with flow cytometry as the reference; this project could not verify that independently. The proposal names the C-NMC 2019 challenge data (10,661 training images from 73 subjects: 7,272 ALL and 3,389 normal; test sets of 1,867 and 2,586 images). Those figures describe the planned database; the measurements below were made on the dataset in the table above.

Getting the images ready (before block 2)

Two preparation steps run first. **Stain normalisation** (Reinhard method in the CIELAB colour space) shifts each image's cell colours towards a common reference, because staining differs between slides and days. The reference was fitted on 20 images (Lab mean 73.3 / 10.3 / −24.9, standard deviation 7.4 / 12.6 / 13.9), and the per-channel gain is limited to a factor between 0.25 and 4 so that an unusual image cannot be stretched wildly. A light **bilateral filter** then reduces noise for the segmentation step.

$$\mathrm{ch}' = \big(\mathrm{ch} - m_{\mathrm{ch}}(I)\big)\, g_{\mathrm{ch}} + m^*_{\mathrm{ch}},$$
$$g_{\mathrm{ch}} = \mathrm{clip}\Big(\frac{s^*_{\mathrm{ch}}}{\max\big(s_{\mathrm{ch}}(I),\, 10^{-3}\big)},\; 0.25,\; 4\Big)$$

Here ch is one of the three Lab channels, $m_{\mathrm{ch}}(I)$ and $s_{\mathrm{ch}}(I)$ are its mean and standard deviation over the cell (foreground) pixels of image I, and m^*, s^* are the reference values above.

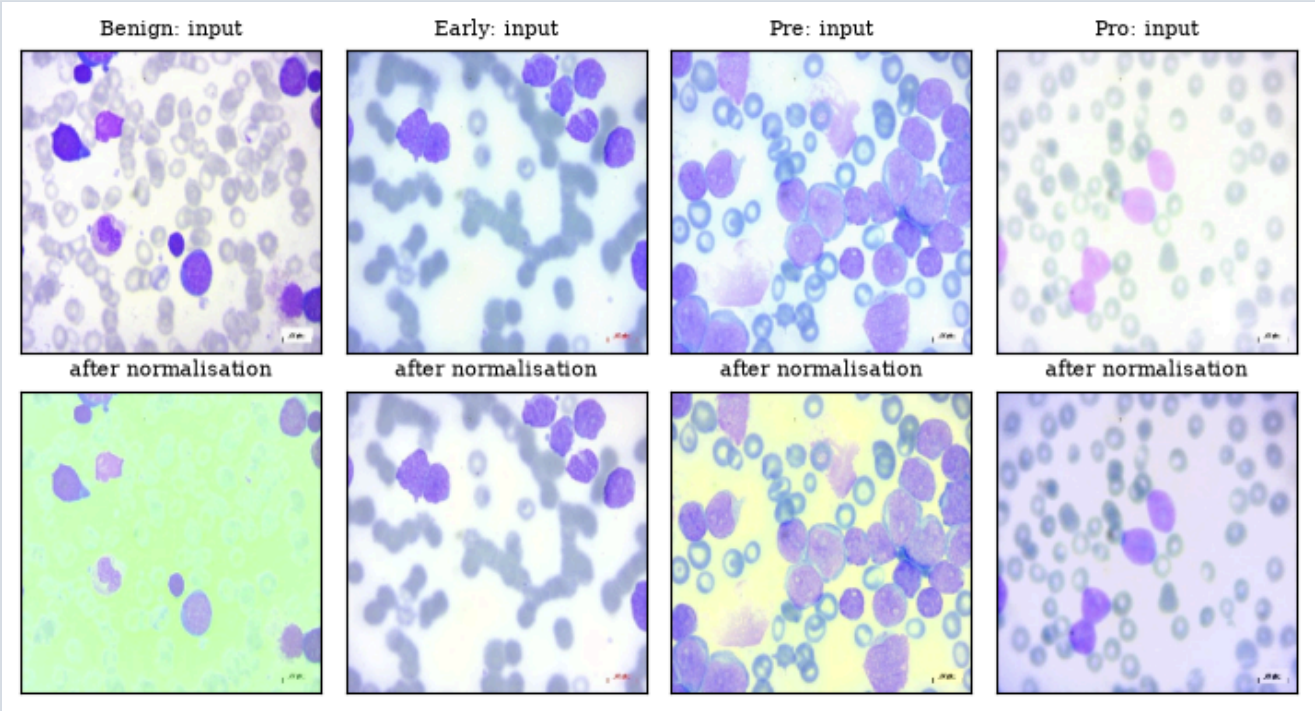


Figure 2. Stain normalisation on one image per class. Top: input. Bottom: after normalisation. Normalisation reduces, but does not remove, differences in background tint between classes.

HONEST LIMIT

The dataset has no patient identifiers. Images are numbered in capture order, and neighbouring numbers look alike (similar background and stain), which is why the evaluation in block 6 keeps whole stretches of numbers apart rather than splitting at random.

Block 2: Proposed HDS Edge Detection

> **IN ONE LINE**
> HDS (Hybrid Diffusion-Steered) smooths away noise inside regions while refusing to smooth across edges, and produces an edge map as a by-product.

The idea in plain words

Imagine ink spreading on wet paper. Noise is like tiny specks of ink: if the ink is allowed to spread inside a flat region, the specks fade. But we do not want ink to leak across the outline of a cell, because that would blur the boundary. **Diffusion filtering** does exactly this on an image: brightness is allowed to flow between neighbouring pixels, and a *conduction coefficient* c decides how freely. Where neighbouring pixels are similar (flat region), c is close to 1 and smoothing is strong. Where they differ sharply (an edge), c drops towards 0 and smoothing stops.

Two classic rules for c exist. **Perona-Malik** reacts to the contrast across an edge and preserves it well. **Total Variation (TV)** flattens regions while keeping boundaries sharp. HDS **combines both**, so the filter keeps the strengths of each:

$$c(s) = \frac{w}{1 + (s/h)^2} + (1 - w)\,\frac{\beta}{\sqrt{s^2 + \beta^2}}, \qquad 0 < c(s) \le 1$$

$$u_{t+1} = u_t + \Delta t \left[\sum_{k \in \{N,S,W,E\}} c\big(|d_k|\big)\, d_k - \lambda\,(u_t - f) \right], \qquad d_k = u_t^{(k)} - u_t$$

Here s is the size of the brightness difference d_k between a pixel and its neighbour in direction k (north, south, west, east), f is the original image and u the image being cleaned. The first part of c is the Perona-Malik rule and the second the Total Variation rule, mixed with weight w. The last term (λ) gently pulls the result back towards the original so that real detail is not erased.

Settings used

w (Perona-Malik share)	h (contrast scale)	β (TV scale)	λ (fidelity)	Δt	Iterations
0.5	0.05	0.05	0.3	0.2	up to 30, stops early when the change falls below 0.0001

The edge map

After cleaning, the average of c over the four neighbours tells how much "smoothing room" each pixel has. The **edge indicator** is 1 minus that average: about 0 in flat regions and close to 1 across strong edges, on a fixed scale (so a quiet image is not artificially stretched to look as edgy as a busy one).

$$E = \mathrm{clip}\Big(1 - \tfrac{1}{4}\sum_{k} c\big(|d_k|\big),\ 0,\ 1\Big)$$

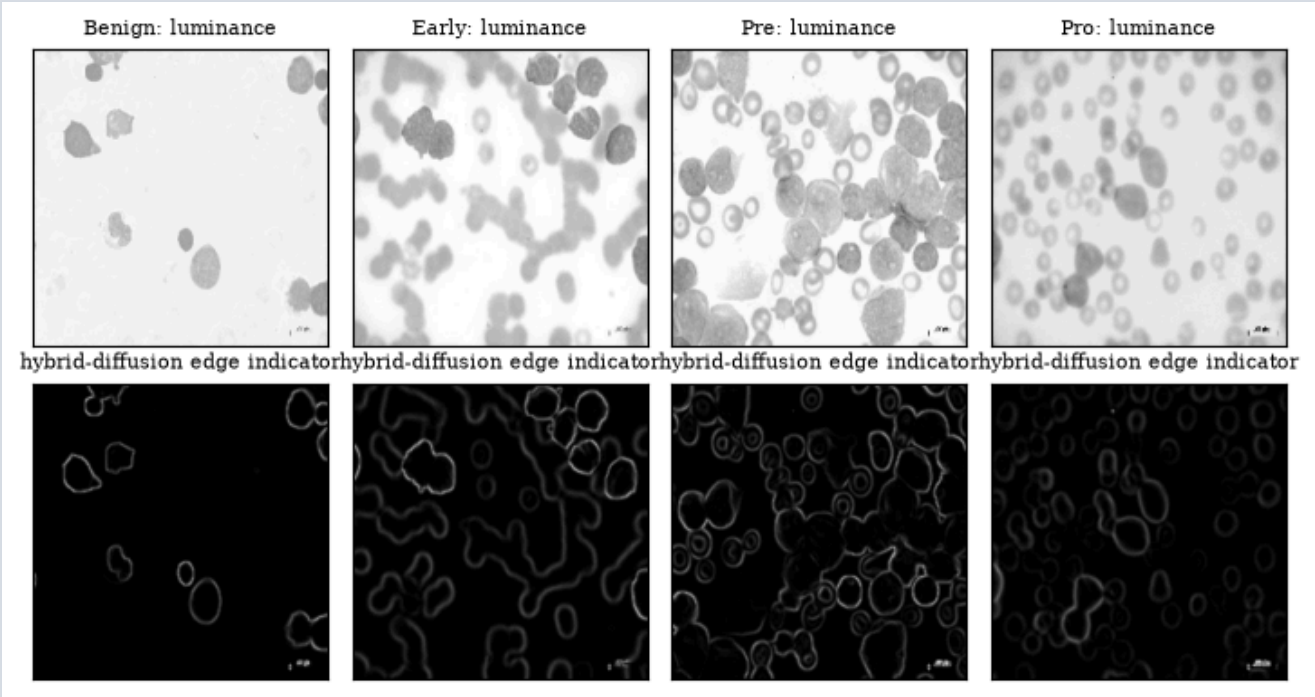


Figure 3. HDS edge indicator. Top: luminance of one image per class. Bottom: the edge indicator (dark = flat, bright = strong edge).

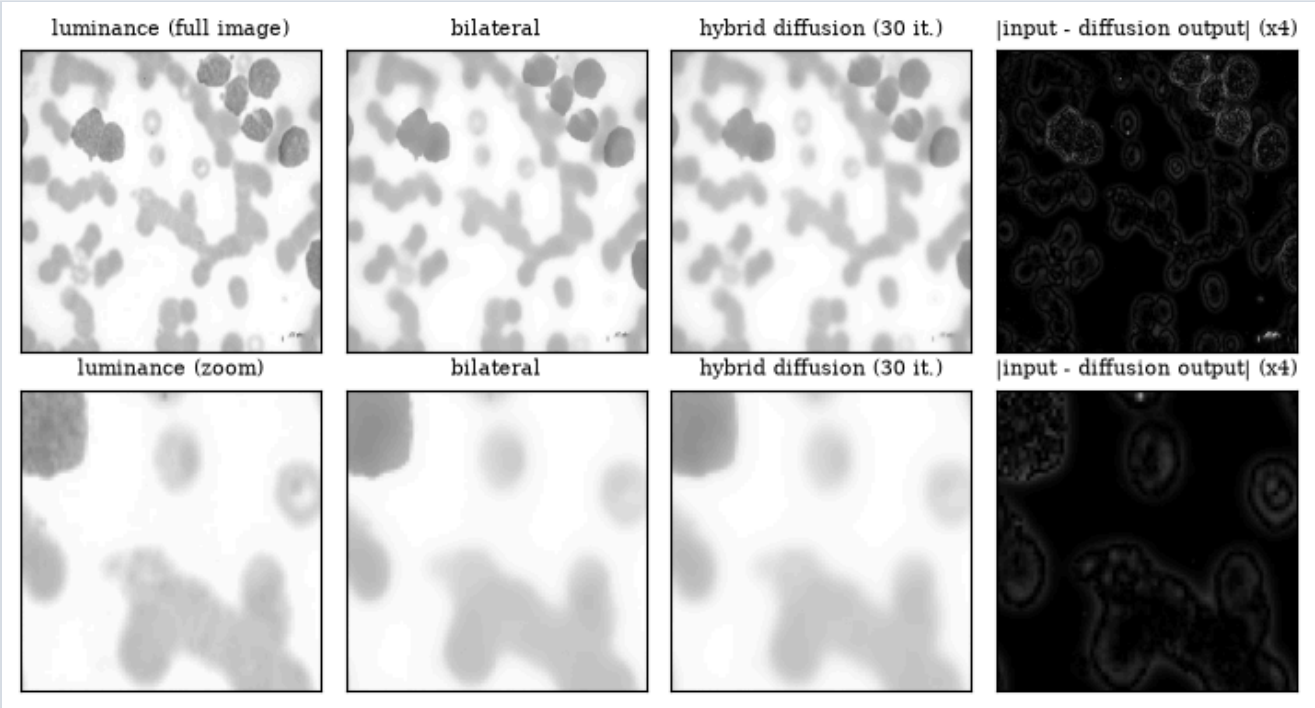


Figure 4. Denoising of one image: input, bilateral filter, stabilised HDS (30 iterations) and the amplified difference. Both keep the cell outlines; the difference image shows where the filter changed the picture.

Making it stable, and how well it works

The original prototype form of the filter has a TV term containing a quantity of size 1/ε (about 10^8 in flat regions). That breaks the stability limit of the update step and makes the filter blow up on this kind of image. Here both rules are rescaled so that c never exceeds 1, and Δt is kept below $1/(4+\lambda) \approx 0.23$ (here $\Delta t = 0.2$). Each update then only mixes neighbouring values, so the output can never leave the range of the input; a 3,000-iteration test stayed finite and inside the input range.

On a synthetic test (clean image plus added noise, three noise levels), the signal quality after filtering, in dB (higher is better):

Noise level σ	Noisy	Original (unstabilised) HDS	Stabilised HDS	Bilateral	Gaussian
0.05	26.1	5.0	37.5	38.2	26.4
0.08	21.9	5.0	34.6	34.0	26.3
0.12	18.4	5.0	31.4	27.7	26.0
Average gain		**−17.2**	**+12.3**	**+11.2**	**+4.1**

It takes about 58 ms per image on a CPU. On the real dataset it was run on all 3,256 images, and its edge map feeds per-cell edge statistics (3 numbers per cell).

HONEST LIMITS

The noise test is synthetic (added noise on a clean image); it has not been validated on real tissue. HDS was designed for speckle noise in ultrasound, while microscopy noise here is additive, which is exactly why the stabilised additive-noise form was needed. Used alone, the HDS edge features reach a balanced accuracy of 0.551 (chance is 0.25); adding them to the other hand-made features raised the combined score from 0.825 to 0.850.

Block 3: APC Three Blood Vessels Segmentation

IN ONE LINE

Find the structures in the image (APC, based on how sharply brightness bends) and outline each white cell (segmentation) so every cell can be examined on its own.

Part A: Adaptive Principal Curvature (APC)

Curvature measures how sharply the brightness surface bends. At every pixel, the second derivatives form a small matrix (the Hessian) whose two eigenvalues are the *largest* and *smallest* principal curvatures ($\kappa_1 \geq \kappa_2$). A thin line or a boundary bends sharply across it; a flat area does not bend at all. APC **combines the two curvatures** instead of relying on brightness change alone:

$$H = \begin{pmatrix} I_{xx} & I_{xy} \\ I_{xy} & I_{yy} \end{pmatrix}_{\sigma}, \qquad \kappa_1, \kappa_2 = \text{eigenvalues of } H \ (\kappa_1 \geq \kappa_2)$$

$$\text{APC} = \text{clip}\Big(\sigma^2 \, g \left[\alpha \, |\kappa_2| + (1-\alpha) \, |\kappa_1|\right], \ 0, \ 1\Big), \qquad \sigma = 2.5, \ \alpha = 0.6, \ g = 12$$

The σ^2 factor makes the response comparable across scales, and the gain is fixed rather than set per image, so an empty field full of noise is not stretched to look as strong as a real cell edge. The proposal also lists central-difference, Sobel and Prewitt derivative operators as variants of how the derivatives are taken; this implementation takes them from a Gaussian-smoothed Hessian.

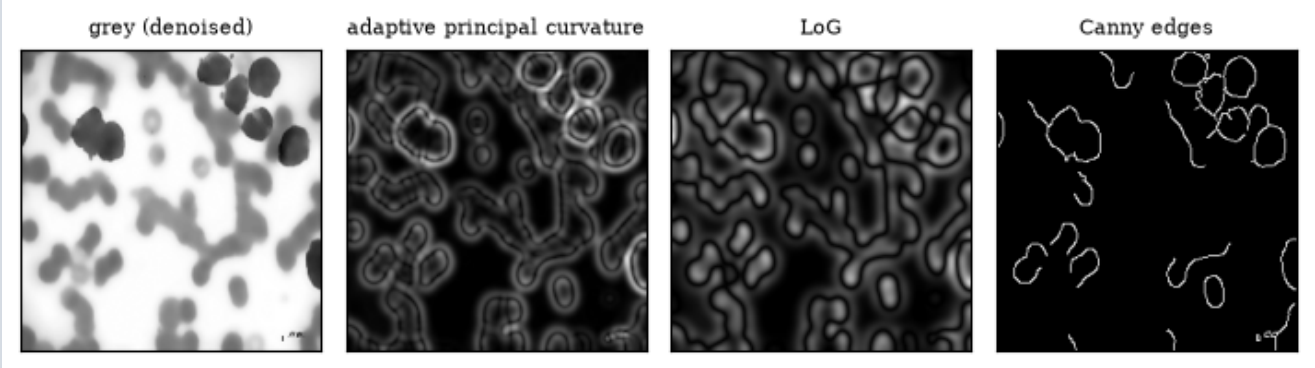


Figure 5. From left: denoised grey image, APC response, LoG (blob) response and Canny edges. APC lights up cell outlines and thin structures.

ABOUT THE NAME "BLOOD VESSELS"

APC was created for vessel-like (ridge) structures, for example in retinal images, which is where the name comes from. A peripheral blood smear contains no blood vessels. In this project the APC response therefore highlights **cell boundaries and thin structures**, and its output is used as an informative structure map, not as a vessel finder.

Part B: Segmenting the cells

To examine cells one by one, each white cell must be outlined. The segmenter works on **colour**: white blood cells are strongly magenta/purple, while red cells and background are pale and grey-blue. On the a* (green-to-red) colour axis this gap is large, so the steps are:

- Smooth the a* channel slightly and threshold it (Otsu threshold, never below 8).
- Remove specks smaller than 120 pixels, fill holes, and apply a light opening.
- Split touching cells with a **watershed** on the distance transform, accepting a split only if every piece is at least 120 pixels.
- Inside each cell, look for a nucleus (the darker part), accepted only when it is clearly different from the cytoplasm and large enough.

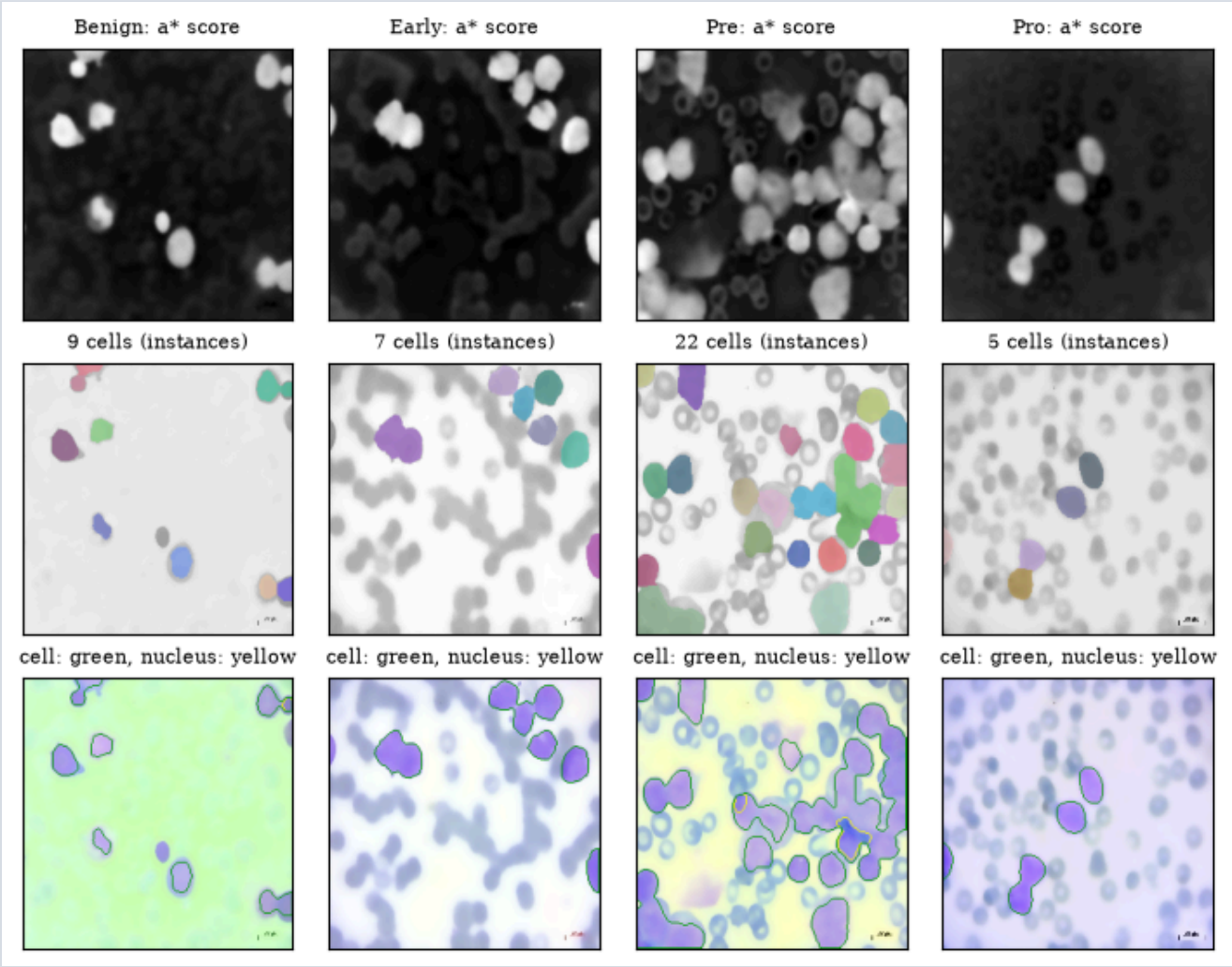


Figure 6. Segmentation for one image per class. Top: smoothed a* score. Middle: individual cells after watershed splitting (random colours). Bottom: cell (green) and nucleus (yellow) outlines.

Result on the database: 36,167 cells in 3,256 images, in about 81 ms per image for segmentation, features and response layers together. Nuclei were reliably separable in only 2,741 cells (7.6%), so nuclear measurements are not used for classification.

From outlines to clean cell crops (the hand-off to blocks 4 and 5)

A check on the database found that the **background colour alone predicts the class** with a balanced accuracy of 0.82 (chance is 0.25), because classes were photographed in different sessions. A classifier could therefore "cheat". To prevent that, every pixel outside the cells (plus a 2-pixel rim) is replaced with one neutral grey and the cells are white-balanced against the slide background. After this step the background probe falls to exactly chance (0.25). Each cell is then cut out as a square crop with a 25% margin, with all other cells blanked out, and resized to 112×112 pixels for the classifier.

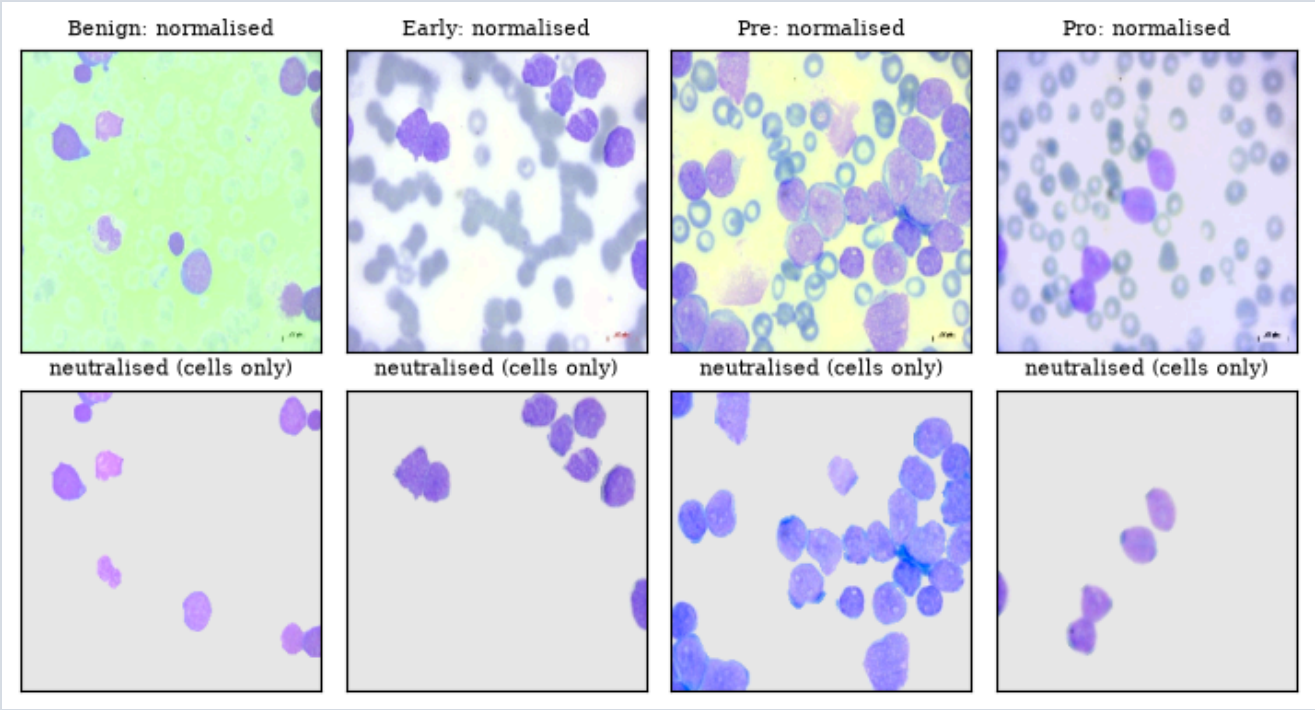


Figure 7. Top: stain-normalised images. Bottom: everything outside the cells replaced by a neutral fill, cells white-balanced.

HONEST LIMITS

No manual segmentation masks come with this dataset, so the Dice, Jaccard and contour-matching scores named in the proposal could not be computed; segmentation quality was checked on example images. Cell colour can still carry session information after white balancing (the mean cell colour alone gives 0.80), so a high accuracy cannot be read as proof that only disease morphology was used.

Block 4: BEEMD Based Pure IMF Selection

> **IN ONE LINE**
> Take the image apart into layers from the finest detail to the coarsest background, then keep only the layers that carry real cell structure.

The idea in plain words

Think of a piece of music separated into treble, mid and bass. **Empirical Mode Decomposition (EMD)** does the same to an image. It finds the fastest-changing pattern in the picture, removes it as the first layer, then finds the fastest pattern left, and repeats. The layers are called **Intrinsic Mode Functions (IMFs)**, ordered from fine to coarse, and what remains at the end is a smooth **residue** (illumination and background). Nothing is lost: all IMFs plus the residue add back to the original image exactly (the largest error measured was 1.2×10^{-7}).

$$I(x,y) = \sum_{k=1}^{K} \mathrm{IMF}_k(x,y) + r_K(x,y), \qquad K = 4$$

Each layer is found by **sifting**: build an upper envelope through the local peaks and a lower envelope through the local valleys, subtract their average, and repeat until the result stops changing (here at most 8 sifts).

$$h_{j+1} = h_j - \tfrac{1}{2}\big(e_j^{\max} + e_j^{\min}\big), \qquad \text{stop when } \frac{\sum (h_j - h_{j+1})^2}{\sum h_j^2} < 0.2$$

$$\mathrm{IMF}_k = \frac{1}{N}\sum_{i=1}^{N} \mathrm{IMF}_k^{(i)}, \qquad N = 8, \;\; \text{each run on } I \pm \eta_i, \;\; \eta_i \text{ of size } 0.2\,\mathrm{std}(I)$$

- **Bidimensional (B):** it works directly on a 2D image rather than on a signal.
- **Ensemble (E):** EMD can be unstable, so the decomposition is repeated 8 times on the image plus small random noise (20% of the image's standard deviation, added in cancelling + / − pairs) and the layers are averaged.
- **Fast:** the envelopes are built with maximum/minimum filters whose window size follows the spacing of the peaks, instead of slow curve fitting. The result is fast and deterministic.

Four IMFs are extracted per image.

Selecting the "pure" IMFs

Not every layer is useful: the finest is mostly pixel noise, and the coarsest is mostly lighting drift. A layer is kept as **pure** if its characteristic spatial period lies between **3 and 48 pixels** (cell-scale structure) and it carries at least **1% of the detail energy**. The kept layers are summed into one "pure-IMF" image.

$$\text{keep IMF}_k \iff 3 \le P_k \le 48 \text{ px} \;\text{ and }\; \frac{E_k}{\sum_j E_j} \ge 0.01, \qquad P_k = \frac{1}{\bar{f}_k}, \;\; E_k = \sum_{x,y} \mathrm{IMF}_k(x,y)^2$$

P_k is the characteristic spatial period of layer k (the inverse of its radial spectral centroid $\bar{f}_k$, in cycles per pixel) and E_k its energy.

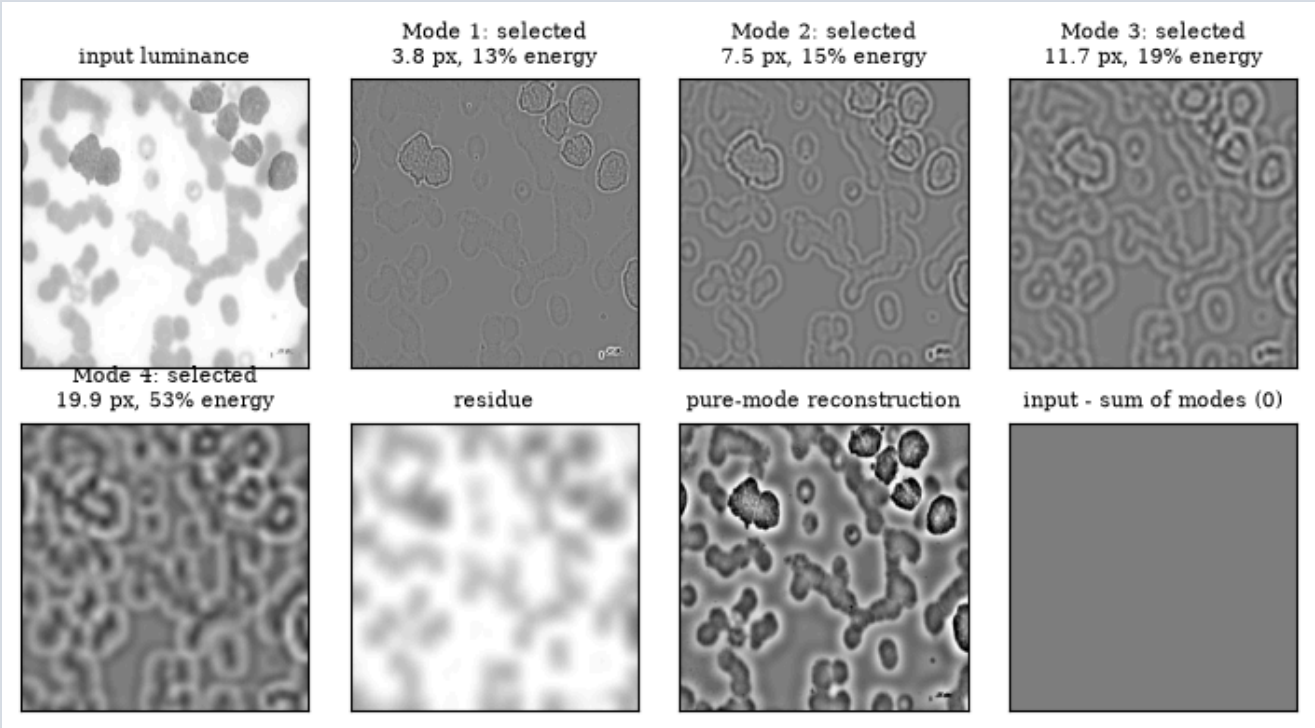


Figure 8. BEEMD on one image: the four IMFs from fine to coarse (each titled with its spatial period, share of detail energy and whether it is kept), the residue, the sum of the selected modes, and the reconstruction difference (zero up to rounding).

Example image	IMF 1	IMF 2	IMF 3	IMF 4
Spatial period (pixels)	3.8	7.5	11.7	19.9
Share of detail energy	12.8%	15.0%	19.5%	52.8%
Kept as pure?	yes	yes	yes	yes

How it is used and what it measured

For each cell, the pure-IMF image is summarised as its mean, spread and edge-rim average, and the energy of each IMF is recorded (7 numbers per cell). BEEMD is the slowest stage, about 0.75 s per image on a CPU. In the feature analysis, the pure-IMF numbers alone reach a balanced accuracy of 0.515 and the IMF energies 0.446 (chance is 0.25). Adding them to shape, texture, APC, LoG and HDS features raised the combined hand-made score from 0.850 to **0.875**.

> **HONEST LIMIT**
> With these thresholds the selection rule almost never rejects a layer: it kept the first IMF in 99.97% of images and the other three in 100%. In this data it acts as a pass-through, not a strong filter. The thresholds (3 to 48 pixels, 1%) are a reasoned choice; the rule can be replaced by changing one function.

Block 5: Classification using FBPTAT NN

> **IN ONE LINE**
> A neural network that turns what was learned about the cells into a decision: Benign, Early, Pre or Pro.

The proposed network: FBPTAT

The proposal describes **FBPTAT** (Fractional Back-Propagation Through Adaptive Time): a **recurrent neural network** trained with **fractional-order gradient descent**. A normal network learns by repeatedly nudging its weights along the first derivative (gradient) of the error. The fractional version uses a derivative of fractional order instead, which is not purely local (it depends on more than the slope at the current point); the cited work (Pu and Wang's fractional-order back-propagation, 2018) reports better global optimisation than classic back-propagation, and the proposal states better results than standard back-propagation through time on system identification, pattern classification and Mackey-Glass time-series prediction. In the block diagram this classifier receives the numbers produced by blocks 2 to 4 (edge, curvature and pure-IMF measurements per cell).

$$\underbrace{w_{t+1} = w_t - \eta \left.\frac{\partial E}{\partial w}\right|_{w_t}}_{\text{ordinary (first-order) update}} \qquad \underbrace{w_{t+1} = w_t - \eta \left. D_w^{\alpha} E\right|_{w_t}}_{\text{fractional-order update}}$$

E is the training error, η the step size and D^{α} a derivative of fractional order α; $\alpha = 1$ gives back the ordinary gradient.

> **IMPORTANT**
> An FBPTAT network was **not implemented or trained** in this project's software, so this document reports **no FBPTAT results**. What was built and measured for this block is the classifier described next, and every performance number in this document refers to it.

The classifier that was built and measured

The principle is the same as in the diagram: cells go in, a class comes out. An image holds several cells, but only the image has a label. This is a **multiple-instance** problem, and the model treats each image as a **bag of cells**:

1. **Encoder.** Each 112×112 cell crop is turned into a list of 384 numbers by **DinoBloom-S**, a vision transformer pre-trained on a large collection of blood-cell images. It is kept **frozen** (not changed), so each cell is converted once and the results are stored.
2. **Attention head.** A small network scores how informative each cell is (the *attention weight*). The weights are normalised to add up to 1 over the cells of the image, so the result does not depend on how many cells there are, and the weighted average of the cells is the *image summary*.
3. **Output.** A final layer converts the summary into four scores: Benign, Early, Pre, Pro.

$$z_k = \mathrm{GELU}(W h_k + b), \qquad a_k = \frac{\exp\{\mathbf{w}^\top [\tanh(V z_k) \odot \sigma(U z_k)]\}}{\sum_{j=1}^{K} \exp\{\mathbf{w}^\top [\tanh(V z_j) \odot \sigma(U z_j)]\}}$$

$$\mathbf{b} = \sum_{k=1}^{K} a_k \, z_k, \qquad \hat{\mathbf{p}} = \mathrm{softmax}(C \, \mathbf{b})$$

h_k: the 384 numbers of cell k; z_k: its 256-number hidden version; a_k: the attention weight of the cell (the weights add up to 1); $\mathbf{b}$: the image summary; $\hat{\mathbf{p}}$: the four class probabilities.

The only trained part is the head, with **165,509 parameters**. It was trained with label smoothing (0.1), class weights that compensate for unequal class sizes, the AdamW optimiser with warm-up and cosine decay, and early stopping on a validation set. A single temperature value, fitted on validation data, then makes the output probabilities honest (calibration).

Attention also makes the decision partly inspectable, because it shows which cells the model relied on for an image.

Other classifiers measured on the same protocol

Classifier	Input	Balanced accuracy
Frozen DinoBloom-S + attention head (headline)	cell crops	**0.968**
Low-rank adaptation (LoRA) of DinoBloom-S, trained on a GPU	cell crops	0.917
Hand-made features (shape, texture, APC, LoG, HDS edge, pure-IMF, IMF energy) + logistic head	31 per-cell statistics	0.875

Two observations follow. First, the features from blocks 2 to 4 carry genuine signal (0.875 against a chance level of 0.25), so those blocks contribute. Second, fine-tuning the encoder did *not* help: the adapted model scored 5 percentage points lower than the frozen one, so the frozen features were not the limiting factor. A classifier that takes the block 2-4 features, such as the proposed FBPTAT network, belongs at exactly this point of the pipeline.

Block 6: Cancer Detection

> **IN ONE LINE**
> The final answer: for an image, the predicted class, how confident the system is, and whether the sample is benign or malignant.

What the system outputs

- **Class** (Benign, Early, Pre or Pro) and a probability for each.
- **Benign vs malignant:** the three malignant probabilities added together.
- **Which cells mattered** (attention) and how many cells were found.

How performance was measured

Accuracy is easy to inflate when images from the same session sit on both sides of the train/test line. Therefore, within each class the images were sorted by capture number and cut into **five consecutive segments**. One segment is the test set, another is the validation set, and training images within 37 numbers of any test or validation image are removed (an *embargo*). Five models are trained, one per fold, and their predictions on unseen images are pooled. Confidence intervals come from resampling groups of 37 consecutive images (90 groups), not single images.

Measure	Result	What it means
Balanced accuracy	0.968 (95% CI 0.955 to 0.978)	Average of the per-class hit rates, so a small class counts as much as a large one.
Macro-F1	0.967	Balance of precision and recall, averaged over classes.
AUROC (one-vs-rest)	0.995	How well the scores rank the right class above the others.
Sensitivity (malignant found)	0.990	Share of Early, Pre and Pro images called malignant.
Specificity (benign cleared)	0.960	Share of Benign images called benign.
Calibration error (ECE)	0.010	Gap between stated confidence and actual hit rate (after temperature scaling).

$$\text{balanced accuracy} = \frac{1}{K}\sum_{c=1}^{K}\frac{TP_c}{TP_c + FN_c}, \qquad \text{sensitivity} = \frac{TP}{TP+FN}, \qquad \text{specificity} = \frac{TN}{TN+FP}$$

$$F_1^{(c)} = \frac{2P_cR_c}{P_c+R_c}, \quad \text{macro-F1} = \frac{1}{K}\sum_{c} F_1^{(c)}, \qquad \text{ECE} = \sum_{b=1}^{15}\frac{n_b}{N}\left|\text{acc}_b - \text{conf}_b\right|, \qquad \hat{\mathbf{p}} = \text{softmax}(\mathbf{z}/T)$$

TP, FN, TN, FP: true/false positives and negatives; P_c, R_c: precision and recall of class c; T: the single temperature fitted on validation data; n_b: images in confidence bin b.

Per-class hit rates are Benign 0.960, Early 0.973, Pre 0.957 and Pro 0.980. Per-fold balanced accuracy ranges from 0.943 to 0.989.

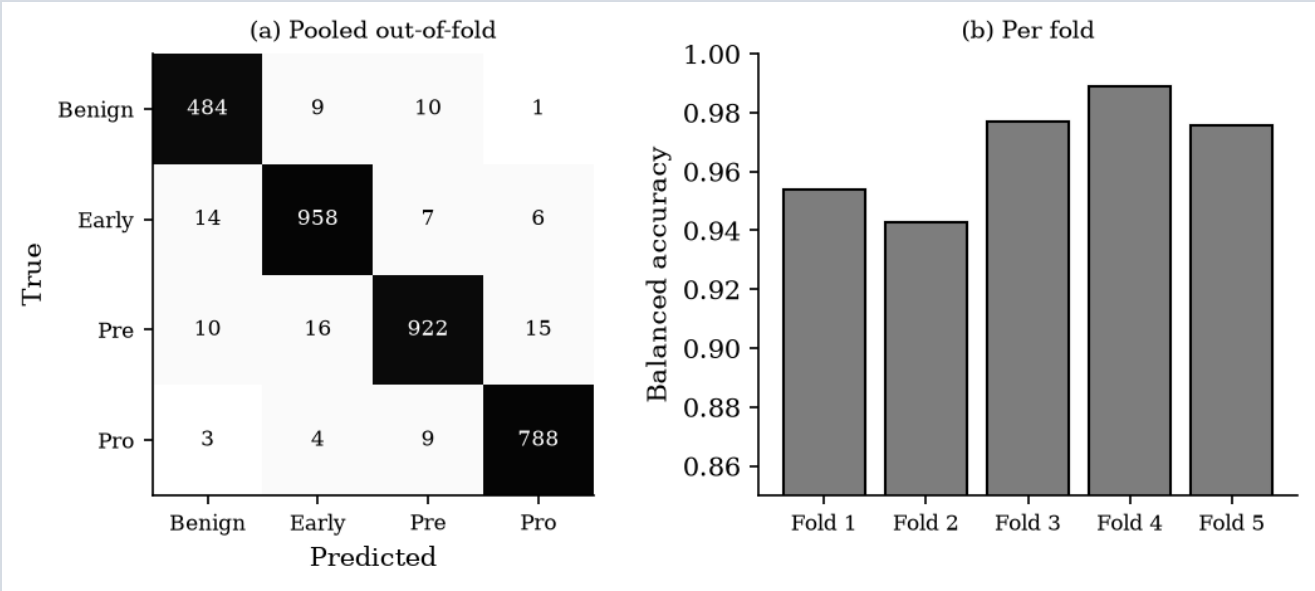


Figure 9. (a) Pooled confusion matrix over all 3,256 images, each tested once by a model that did not train on it (rows: true class, columns: prediction). (b) Balanced accuracy of each fold.

WHAT THIS RESULT IS, AND IS NOT

This is a measurement on one public dataset under a protocol designed to limit leakage. Capture-order groups stand in for patients because no patient IDs exist, so independence between train and test cannot be guaranteed. The proposal's final step, testing on real patient samples, was not done. The system is a research prototype and a screening aid; it is not a diagnostic device, and its accuracy should not be quoted as accuracy for patients in general.

Block 7: GUI (built separately)

> **IN ONE LINE**
> The front end that lets a person use the whole pipeline without touching code.

The GUI is built and maintained separately from the analysis software, so its screens are not described in this document. What can be stated is how it connects to the system. For a single smear image the back end can provide:

The user supplies	The system can return
One blood-smear image (224×224, as in the database)	the number of cells found and their outlines; the stage views (normalised, HDS edge, APC, BEEMD layers, segmentation, cleaned crops); the four class probabilities, the predicted class and the benign-versus-malignant probability; the attention weight of every cell (which cells drove the decision).

The same pipeline is available from the command line as `predict.py <image>`, which prints the class, the probabilities and the per-cell attention, and can use one fold model or an average of all five. It needs the stored model weights and the DinoBloom-S weight file (`data/weights/DinoBloom-S.pth`).

> **ALWAYS SHOWN WITH THE RESULT**
> "Research prototype, not for clinical diagnosis." A single image gives one answer; confirmation of leukemia needs bone-marrow examination and laboratory tests.

How the blocks connect in practice

Figure 1 draws one line from block 1 to block 6. In the working system there are two routes that start from the same cleaned images, and this explains the numbers in blocks 2 to 5:

Route	Path	Balanced accuracy
Headline route	Database → stain normalisation → cell segmentation (block 3) → neutral background and cell crops → frozen DinoBloom-S + attention head (block 5) → detection (block 6)	0.968
Interpretable route	Database → HDS edge map (block 2) + APC and LoG responses (block 3) + BEEMD pure-IMF layers (block 4) → per-cell measurements with shape and texture → simple classifier (block 5) → detection (block 6)	0.875

The interpretable route is the one in which blocks 2, 3 and 4 directly feed a classifier, and it is where a network such as FBPTAT would be plugged in. Its numbers are lower than those of the headline route, but every input is an explicit measurement that can be examined.

What can and cannot be claimed

CAN BE CLAIMED (MEASURED)

- A complete pipeline from smear image to four-class decision runs on all 3,256 images.
- HDS is stabilised and, on synthetic noise, gives an average gain of +12.3 dB (original form: −17.2 dB).
- APC, LoG, HDS-edge and BEEMD features each carry signal, and together reach 0.875 balanced accuracy.
- The frozen encoder with an attention head reaches 0.968 (CI 0.955 to 0.978), with sensitivity 0.990 and specificity 0.960 for malignant versus benign.

CANNOT BE CLAIMED

- Any result for an FBPTAT recurrent network (not implemented).
- Dice, Jaccard or contour scores for segmentation (no ground-truth masks).
- That APC finds blood vessels in smears, or that pure-IMF selection filters strongly (it keeps almost every mode).
- Accuracy on real patients, other laboratories, or the C-NMC dataset.
- Diagnostic use.

Additional Section: Results, Plots and Outputs Explained

This section collects the measured outputs of the project: the plots, the tables and the actual predictions. For each one it says **what it shows** (how to read the axes), **what we see** (the numbers) and **what it means** (including what it does *not* prove). All numbers come from runs of the software in this repository on the 3,256-image database. Unless stated otherwise, the headline model is the frozen DinoBloom-S encoder with the attention head (block 5), tested with the five-fold protocol of block 6.

> **HOW THE GROUPS OF RESULTS ARE ORDERED**
>
> R1 to R3: what the data look like, and the shortcuts hidden in them. R4 to R6: how the pre-processing blocks (HDS, APC, BEEMD) perform and what they cost. R7 to R13: how the classifier trained and how well it works. R14 and R15: what the classifier relies on, and how evaluation design changes the score. R16: the fine-tuning experiment run on a GPU. R17: real predictions for individual images.

R1. The data: classes and example images

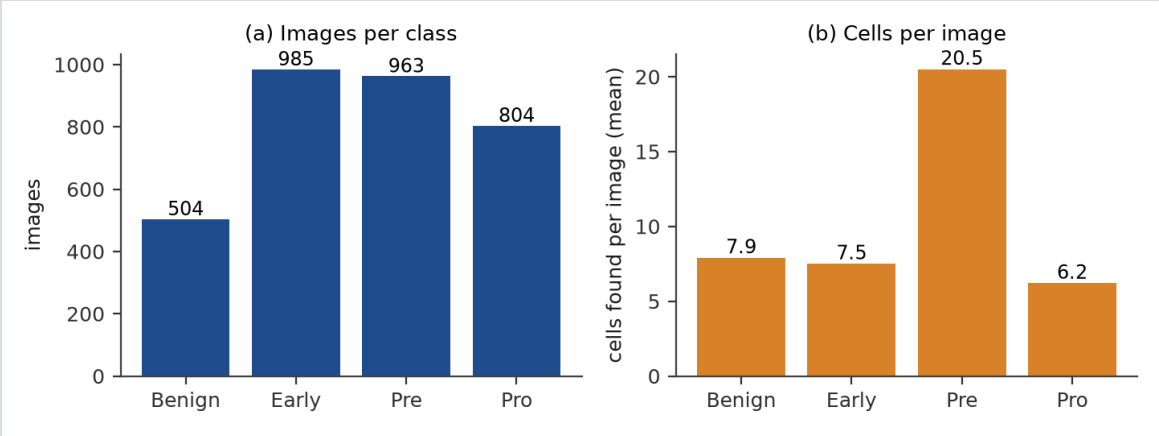


Figure R1a. (a) Number of images per class. (b) Mean number of cells the segmenter found per image.

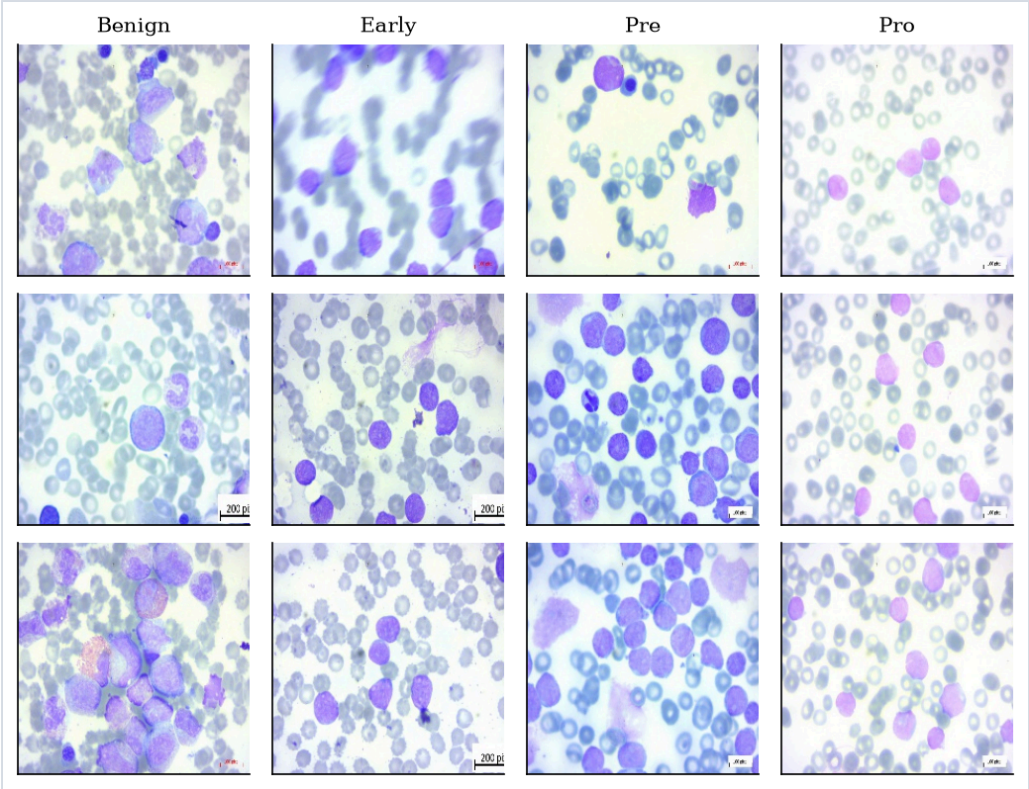


Figure R1b. Three example images per class (224×224 pixels as supplied). Columns: Benign, Early, Pre, Pro.

> **What it shows.** The size of each class and what the raw inputs look like.
>
> **What we see.** Early is the largest class (985 images) and Benign the smallest (504), so the classes are unequal. Pre images contain about 20.5 cells on average, between two and three times as many as Benign (7.9), Early (7.5) or Pro (6.2). In the examples, the background tint and lighting differ from class to class, and the scale bars in the corners differ in style (red, black or white).
>
> **What it means.** Unequal classes are why the project reports *balanced* accuracy and weights the loss. Differences in background and scale bar mean the classes differ in *how they were photographed*, not only in biology. That is exactly what R2 and R3 test.

R2. Neighbouring images look alike (capture-order structure)

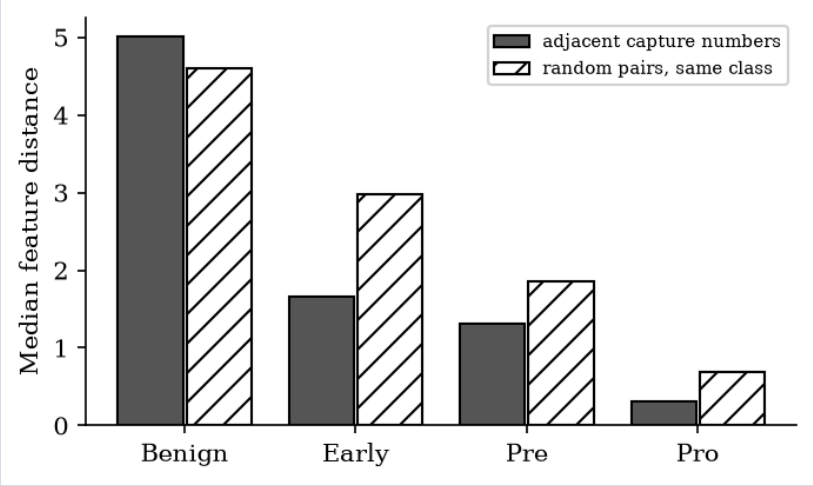


Figure R2. Median colour distance of the image background between images with adjacent capture numbers (dark bars) and between random pairs of the same class (hatched bars).

What it shows. Images are numbered in the order they were captured. For each class, the plot compares how similar the background colour is for images with *neighbouring numbers* against images picked at random (smaller bar = more similar).

What we see. Neighbours are clearly more alike than random pairs for Early (1.66 against 2.98), Pre (1.31 against 1.85) and Pro (0.30 against 0.69). Benign shows no such effect (5.00 against 4.60).

What it means. Consecutive numbers behave like images from the same slide or session, so a random split would put near-duplicates of a test image into training and inflate the score. This is the reason for the contiguous five-fold protocol with an embargo of 37 numbers (block 6). It is evidence about capture order, not about patients, because no patient IDs exist.

R3. Shortcut audit: what could a classifier cheat with?

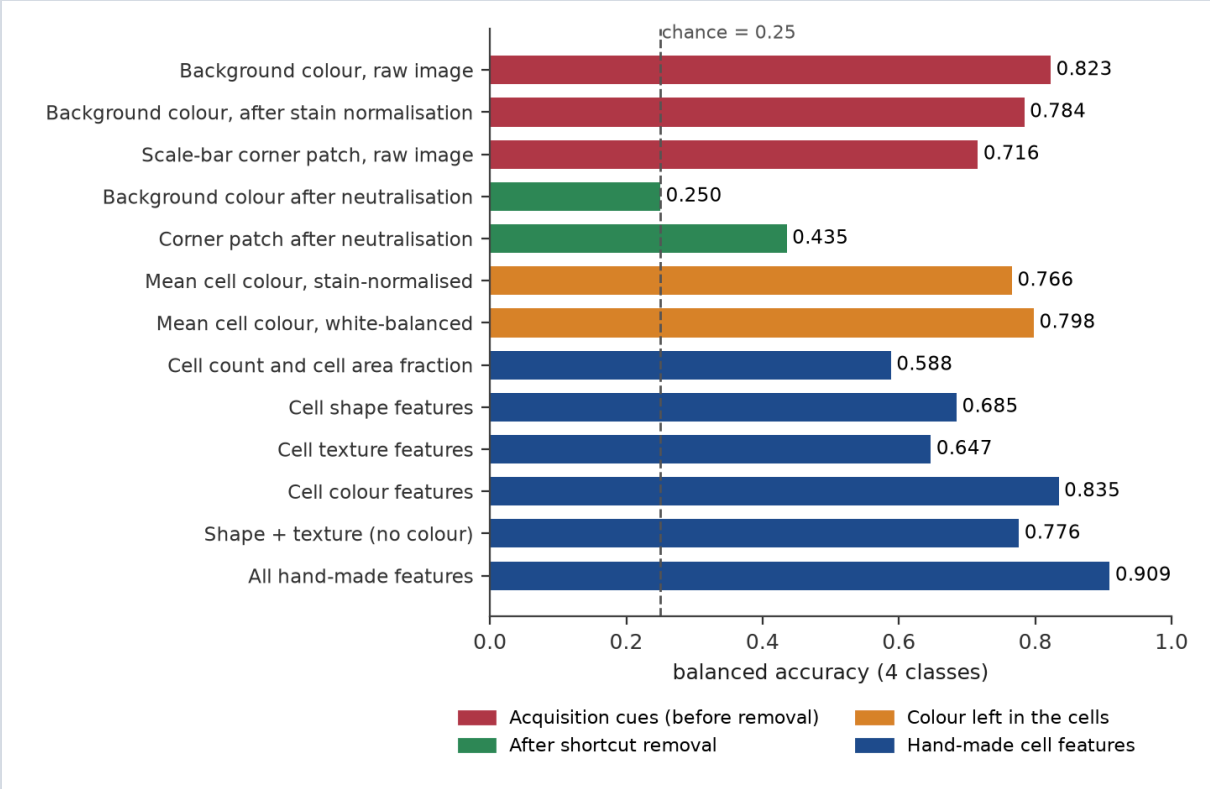


Figure R3. Balanced accuracy of a random-forest classifier that sees only the named quantity (five folds grouped over 37-image capture blocks, no embargo; three seeds; chance = 0.25).

What it shows. Each bar answers: "if a classifier saw *only* this quantity, how well could it guess the class?" Red bars are acquisition cues that have nothing to do with the cells. Green bars are the same cues after the neutralisation step of block 3.

What we see. Background colour alone predicts the class with 0.823 (raw) and 0.784 (after stain normalisation), and the scale-bar corner patch with 0.716, against a chance level of 0.25. After neutralisation the background carries no information (0.250) and the corner patch drops to 0.435. The colour of the cells themselves is still informative: 0.766 (stain-normalised) and 0.798 (white-balanced).

What it means. The shortcut removal in block 3 works for the background, which is why the headline model is given only isolated cells. But cell colour can still reflect the session as well as the disease, so a high accuracy cannot be read as proof that only disease morphology was used. The blue bars are separate hand-made baselines from the audit tool; they include colour and cell-density features, so they differ from the stage analysis in R6, which excludes colour.

R4. Block 2 (HDS): denoising benchmark

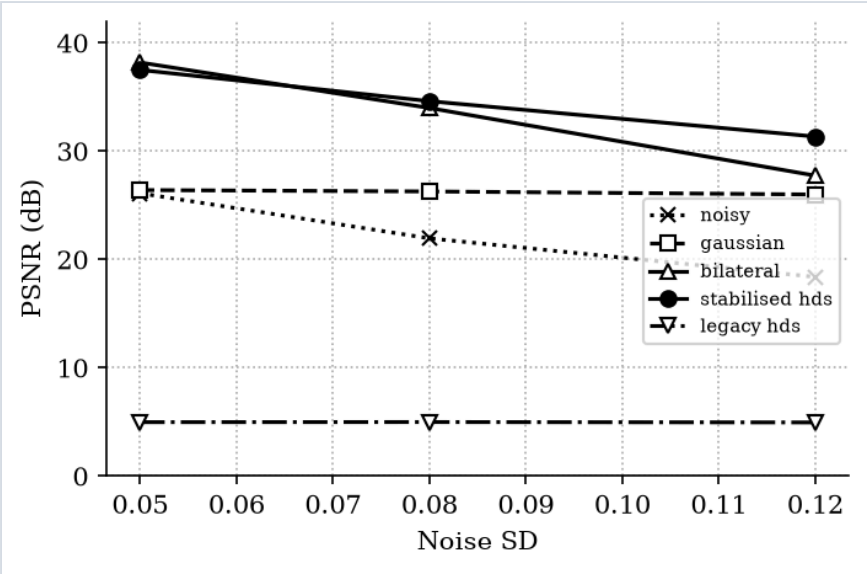


Figure R4. Peak signal-to-noise ratio (dB, higher is better) after filtering a clean image with added Gaussian noise of standard deviation 0.05, 0.08 and 0.12.

> **What it shows.** How well each filter recovers a known clean image from a noisy copy. The dotted line is the noisy image itself, so a useful filter must sit above it.
>
> **What we see.** Stabilised HDS (filled circles) improves on the noisy image by 11 to 13 dB at every noise level: 37.5, 34.6 and 31.4 dB against 26.1, 21.9 and 18.4 dB. The bilateral filter is slightly better at the lowest noise (38.2 dB) but falls behind at the highest (27.7 dB against 31.4 dB). Gaussian smoothing stays near 26 dB. The original, unstabilised scheme (inverted triangles) sits at about 5 dB, below the noisy input: it diverged.
>
> **What it means.** The stabilisation is what makes HDS usable, and it is competitive with the best classical filter. The benchmark is synthetic (added noise on a clean image), so it does not prove an advantage on real smears.

R5. Cost of each block

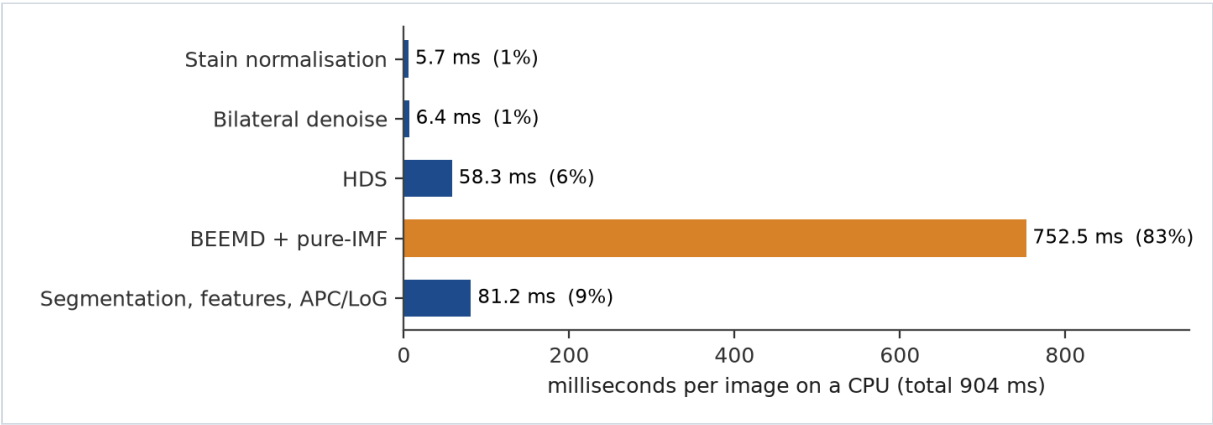


Figure R5. Processing time per image on a CPU for each stage.

> **What it shows.** The time each stage takes for one image (milliseconds), measured on a CPU.

> **What we see.** The whole chain takes about 904 ms per image. BEEMD (with pure-IMF selection) takes 752 ms, which is 83% of the total. HDS takes 58 ms, segmentation with features and the APC/LoG layers 81 ms, and stain normalisation and bilateral denoising about 6 ms each.
>
> **What it means.** The ensemble decomposition in block 4 is the expensive part. Across 3,256 images it adds about 41 minutes on one CPU, whereas the other stages together take about 8 minutes. If speed matters (for example in the GUI), block 4 is where to save time.

R6. What blocks 2, 3 and 4 add to a simple classifier

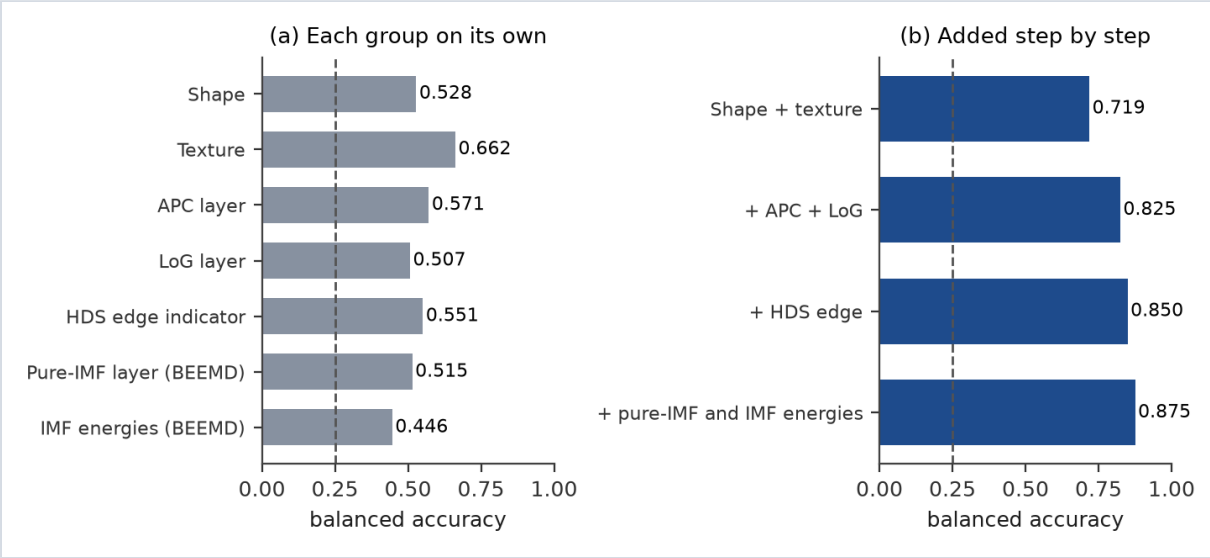


Figure R6. Balanced accuracy of a logistic-regression classifier on image-mean hand-made features (contiguous folds, embargo 37, three fold rotations; chance = 0.25). (a) Each feature group alone. (b) Groups added one after another.

> **What it shows.** The "interpretable route" from the table in "How the blocks connect": the numbers measured from blocks 2 to 4 are fed into a simple classifier. Panel (a) tests each group alone; panel (b) tests the cumulative combination.
>
> **What we see.** Every group alone beats chance, but none is strong: texture 0.662, APC layer 0.571, HDS edge 0.551, pure-IMF layer 0.515, shape 0.528, LoG 0.507, IMF energies 0.446. Combined they climb in steps: shape and texture 0.719, adding APC and LoG 0.825, adding the HDS edge 0.850 and adding the pure-IMF layer and IMF energies **0.875**.
>
> **What it means.** The edge, curvature and decomposition blocks each contribute: adding APC and LoG gave about 10.6 points, the HDS edge about 2.5 points and the BEEMD features about 2.6 points. The individual gains are small and come from different groups, so they are best read as complementary. This route stays below the frozen-encoder route (0.968), but every input is an explicit, inspectable measurement.

R7. How the classifier head trained

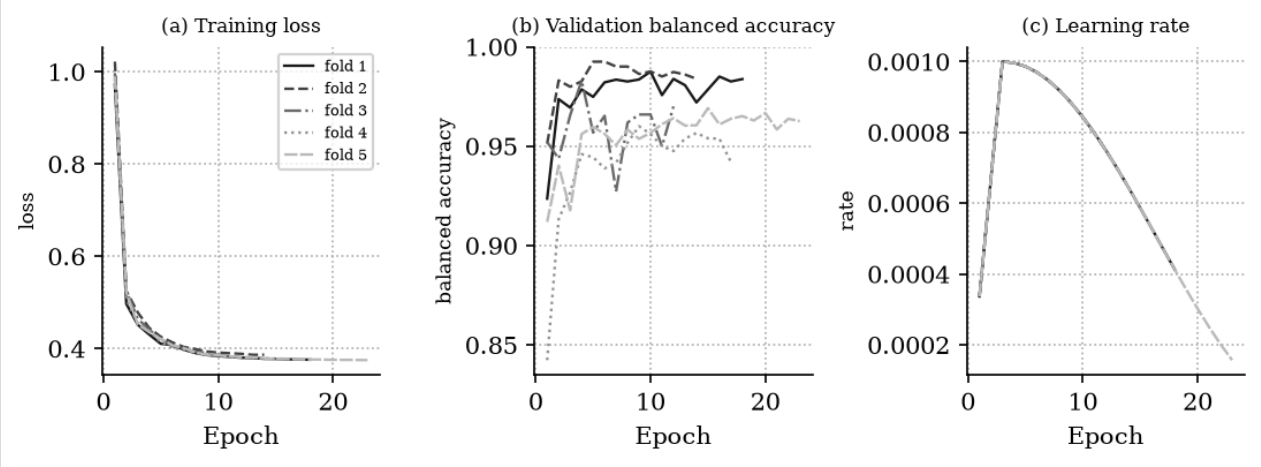


Figure R7. Frozen-encoder attention head, one line per fold. (a) Training loss. (b) Validation balanced accuracy. (c) Learning rate.

$$\mathcal{L} = -\sum_{c=1}^{4} w_c\, \tilde{y}_c\, \log \hat{p}_c, \qquad \tilde{y}_c = (1-\varepsilon)\, y_c + \frac{\varepsilon}{4}, \quad \varepsilon = 0.1, \qquad \min \mathcal{L} \approx 0.35$$

What it shows. How learning progressed epoch by epoch: (a) the error on training images, (b) accuracy on the validation images that were held out for choosing when to stop, (c) the learning rate.

What we see. The loss falls from about 1.0 to about 0.4 within the first five epochs and then flattens near 0.37. Validation accuracy rises quickly to a band between about 0.94 and 0.99 (depending on the fold) and then wobbles within it. The learning rate warms up over about three epochs to 0.001 and then decays along a cosine curve. Training stopped after 18, 14, 12, 17 and 23 epochs in the five folds, when validation macro-F1 had not improved for eight epochs.

What it means. The loss cannot reach zero because of label smoothing (the targets are 0.925 for the true class and 0.025 for the others), and the lowest value possible is about 0.35, so 0.37 means the head has essentially converged. Reaching a plateau within a few epochs fits a small head on fixed features. The wobble in (b) is the noise of a validation set of about 650 images, and it is why the weights of the best epoch, not the last, are kept.

R8. Reading the confusion matrix (Figure 9)

What it shows. Figure 9 in block 6 counts, for the 3,256 images (each tested by a model that never trained on it), what the model answered for each true class.

What we see. The diagonal holds 3,152 correct answers; 104 images (3.2%) were wrong. The three most common mistakes are Pre called Early (16), Pre called Pro (15) and Early called Benign (14). Pre is the hardest class (recall 0.957, 41 errors), Pro the easiest (0.980, 16 errors). Only 1 Benign image was called Pro and 3 Pro images were called Benign.

What it means. Mistakes mostly fall between *neighbouring maturation stages*: Pre sits between Early and Pro, which is where a human would also expect confusion, because the stages form a continuum. The benign-versus-malignant decision is easier than the four-way one: of the 2,752 malignant images, 27 were called benign (sensitivity 0.990), and 20 of the 504 benign images were called malignant (specificity 0.960). The full list of the 104 wrong images is in the paper's appendix.

R9. ROC and precision-recall curves

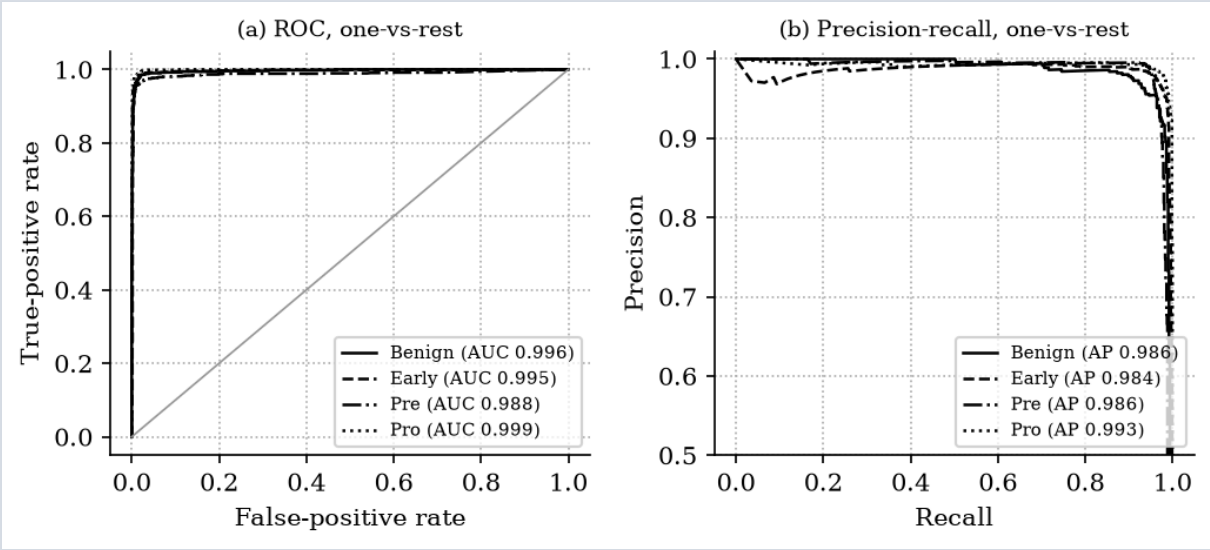


Figure R9. One-versus-rest curves on the pooled out-of-fold predictions. (a) ROC. (b) Precision-recall.

What it shows. Each curve treats one class as "positive" and the other three as "negative". The ROC curve plots the share of true positives found (vertical) against the share of false alarms (horizontal) as the decision threshold sweeps; a curve hugging the top-left corner is ideal. The precision-recall curve plots how many flagged images are correct (precision) against how many true ones are found (recall).

What we see. The areas under the ROC curves are 0.996 (Benign), 0.995 (Early), 0.988 (Pre) and 0.999 (Pro). The average precision is 0.986, 0.984, 0.986 and 0.993. All curves run along the top-left edge and drop only at the far right.

What it means. The scores rank the right class above the others almost always, so errors are concentrated in a few borderline images, not spread thinly. Pre is again the weakest class and Pro the strongest, in agreement with R8. The values are very high on this dataset; they describe this dataset under this protocol and are not a promise for other laboratories.

R10. Calibration: can the percentages be trusted?

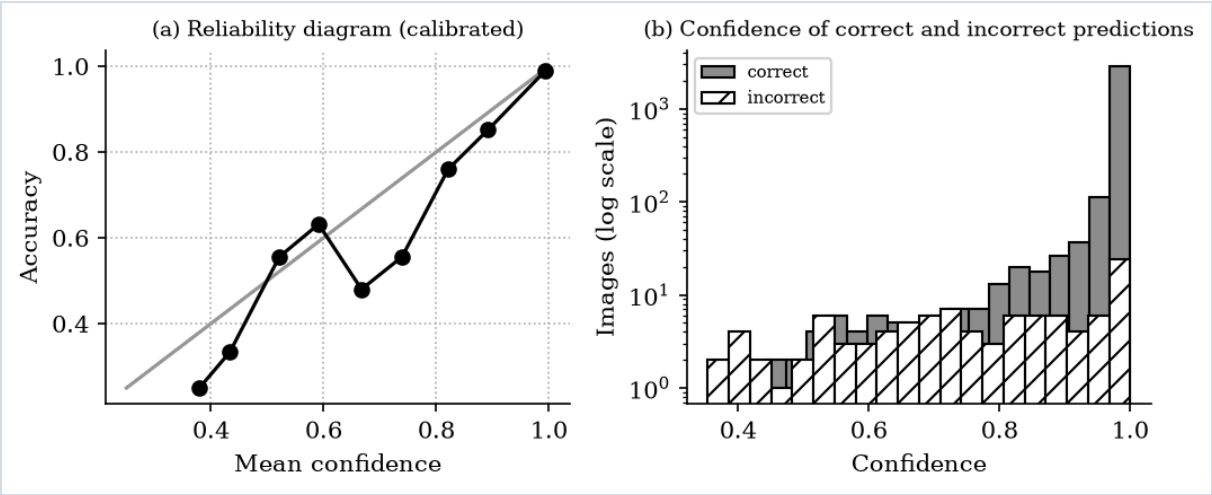


Figure R10. (a) Reliability diagram of the temperature-scaled probabilities. (b) Confidence of correct and incorrect predictions (log scale).

What it shows. (a) Predictions are grouped by the confidence the model stated; the vertical axis shows how often each group was actually right. A perfectly honest model lies on the diagonal. (b) How many correct and incorrect predictions fall at each confidence level.

What we see. Before calibration the raw probabilities were under-confident (mean error 0.121 per fold, fitted temperature between 0.24 and 0.55, always below 1). After one-parameter temperature scaling the mean per-fold error fell to 0.021 and the pooled error is 0.010. The median confidence is 0.998 for correct predictions and 0.811 for wrong ones, and almost all high-confidence predictions are correct.

What it means. Calibrated confidence is informative: a prediction at 0.99 is nearly always right and a wrong one tends to carry lower confidence. The curve in (a) wiggles in the middle because only a small number of images lie in those confidence bins, so it should not be over-read. The temperature was fitted on one dataset and must be re-checked before it is relied on elsewhere. Note that the raw softmax output (for example from a script that does not apply the fitted temperature) understates confidence.

R11. Accuracy depends on how many cells were found

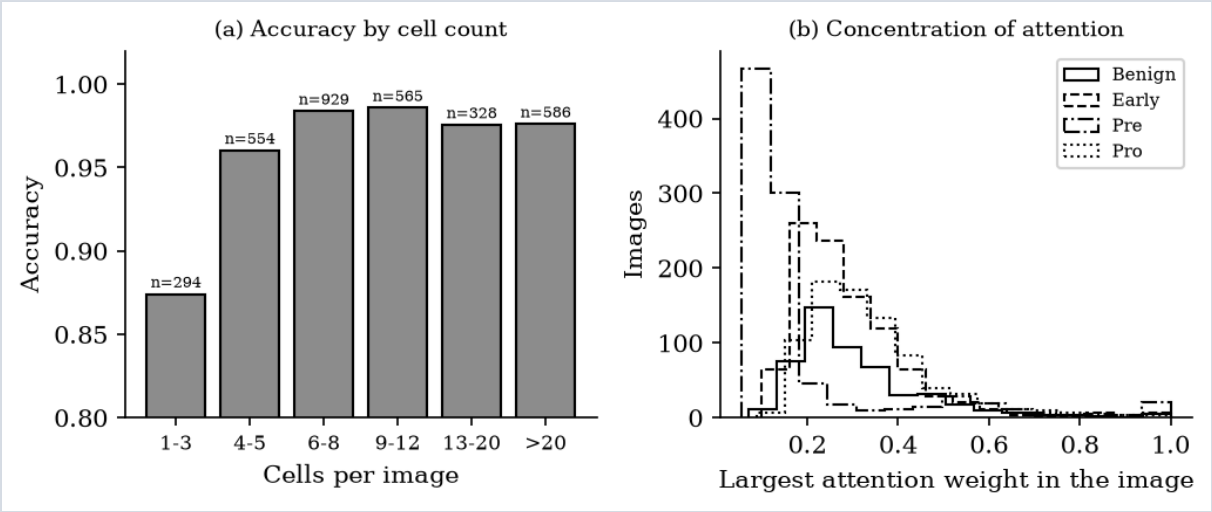


Figure R11. (a) Accuracy by number of cells found in the image (n = number of images). (b) Distribution of the largest attention weight in an image, by class.

> **What it shows.** (a) Whether images with few cells are harder. (b) How strongly the attention head focuses on a single cell, per class (1.0 would mean all weight on one cell).
>
> **What we see.** Images with one to three cells are classified correctly in 87.4% of cases (n = 294). With four to five cells it is 96.0%, and with six or more it is 97.6% to 98.6% (n = 929 for six to eight cells, 565 for nine to twelve). The median largest attention weight is 0.24, about 1.9 times what uniform attention would give. Pre images, which contain many cells, have the smallest maxima.
>
> **What it means.** More cells give the model more evidence to pool, and segmentation may also miss cells in sparse images, although this was not checked image by image. The attention is selective without relying on one cell. For the GUI this suggests flagging results from images with very few detected cells as less reliable.

R12. Which cells did the model look at? (attention)

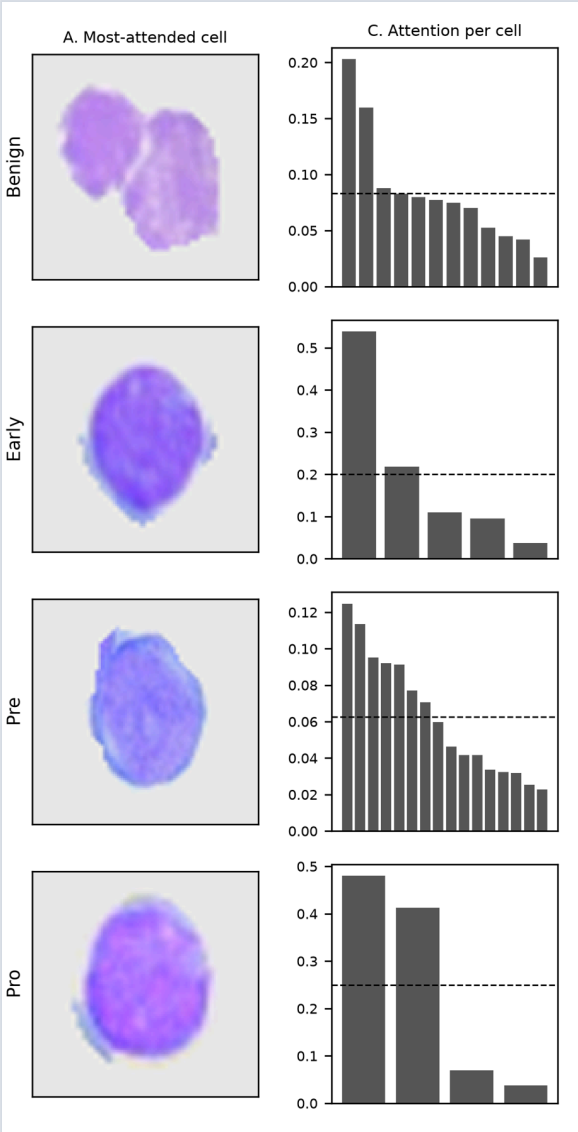


Figure R12. One correctly classified test image per class (fold 1). Left: the most-attended cell. Right: attention weight of every cell in the image, sorted, with the uniform weight as a dashed line.

> **What it shows.** Inside one image, the bars give the share of the decision carried by each cell (the weights add up to 1). The dashed line is what equal attention would give.
>
> **What we see.** In the Early and Pro examples a few cells dominate (in the Early example the top cell carries more than half of the weight). In the Pre example, which has many cells, the weights are spread more evenly. Over the 584 test images of fold 1 with at least three cells, the normalised entropy of the attention was 0.88 on average (1 would be uniform), the largest weight was 2.25 times the uniform weight, and the correlation of attention with cell area was weak and inconsistent (mean Spearman 0.28, SD 0.44).
>
> **What it means.** The head is mildly selective and does not simply favour big cells. Attention shows what the model *weights*; it does not prove those cells are causally important for the disease. A gradient-based saliency method (Grad-CAM) was also tried and failed its sanity checks, so it was not used.

R13. The embedding map (t-SNE)

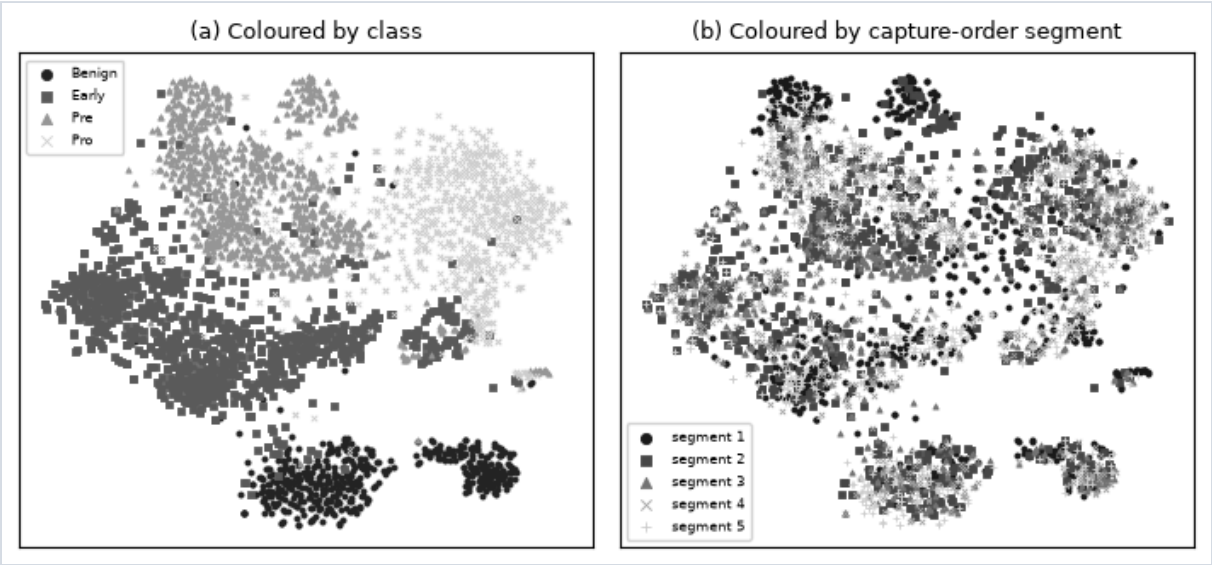


Figure R13. Two-dimensional t-SNE of image embeddings from the frozen encoder. (a) Coloured by class. (b) Coloured by capture-order segment (the fold in which the image is tested).

> **What it shows.** Each dot is an image. The 384-number embeddings were averaged over the cells of the image, reduced to 50 numbers and squeezed into two dimensions so that similar images land close together.
>
> **What we see.** In (a) the Benign images form compact groups apart from the rest, while Early, Pre and Pro occupy neighbouring regions that overlap in places. In (b) images of all five capture-order segments are mixed across most regions. The silhouette coefficient (1 = perfectly separated clusters) is only 0.08 for the class labels and −0.02 for the segment labels.
>
> **What it means.** The classes are not tight, well-separated clusters in the full space, which agrees with the confusions between neighbouring stages. The map does not place the capture-order segments in separate regions, which is mildly reassuring. A two-dimensional picture cannot rule out session information in other directions of the embedding space.

R14. What does the classifier rely on? (input ablation)

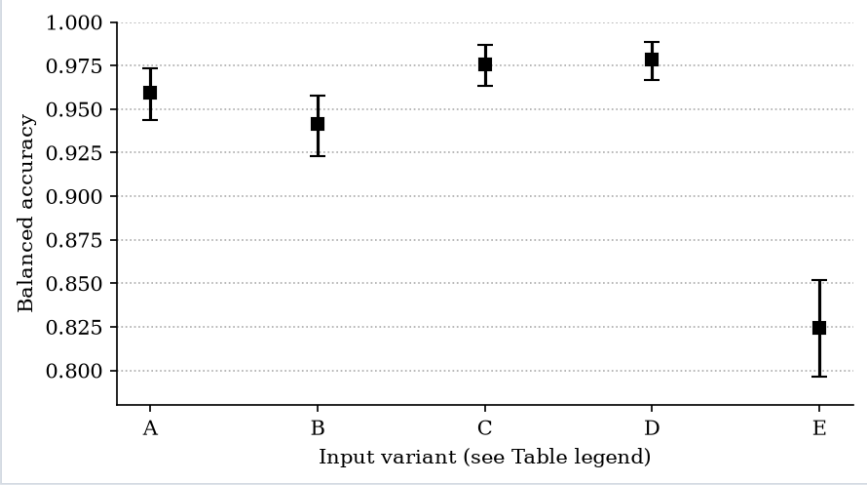


Figure R14. Balanced accuracy (with 95% intervals over session groups) for five different inputs.

Variant	Input given to the model	Balanced accuracy (95% CI)	Difference from A
A	Colour cell crops, background neutralised (reference)	0.959 (0.944 to 0.973)	–
B	Same, converted to grey (no stain colour)	0.940 (0.923 to 0.958)	−0.018
C	Stain-normalised image, background visible	0.975 (0.964 to 0.987)	+0.016
D	Raw image, background visible	0.978 (0.967 to 0.988)	+0.019
E	Hand-made shape, texture and curvature numbers only (no pixels)	0.825 (0.797 to 0.852)	−0.135

What it shows. The same classifier is run on five different versions of the input to see what its accuracy depends on. This uses the faster probe (cell embeddings averaged, then a logistic regression) with three fold rotations, so variant A (0.959) is not the same number as the headline model (0.968, attention head).

What we see. Removing colour (B) costs 1.8 points. Letting the model see the background (C, D) *raises* accuracy by 1.6 to 1.9 points, and these differences are statistically clear (Holm-adjusted p = 0.004, the smallest value possible with 1,000 resamples). The hand-made numbers alone (E) reach 0.825, well below every encoder-based variant.

What it means. Colour carries real signal, which may be biology, staining or session. The background gives extra accuracy that has nothing to do with the cells, which is why the headline model is deliberately not given it: the neutralised setting is the stricter, more honest one. The foundation-model features beat the hand-made ones by about 13 points.

R15. Why the evaluation is built the way it is (leakage analysis)

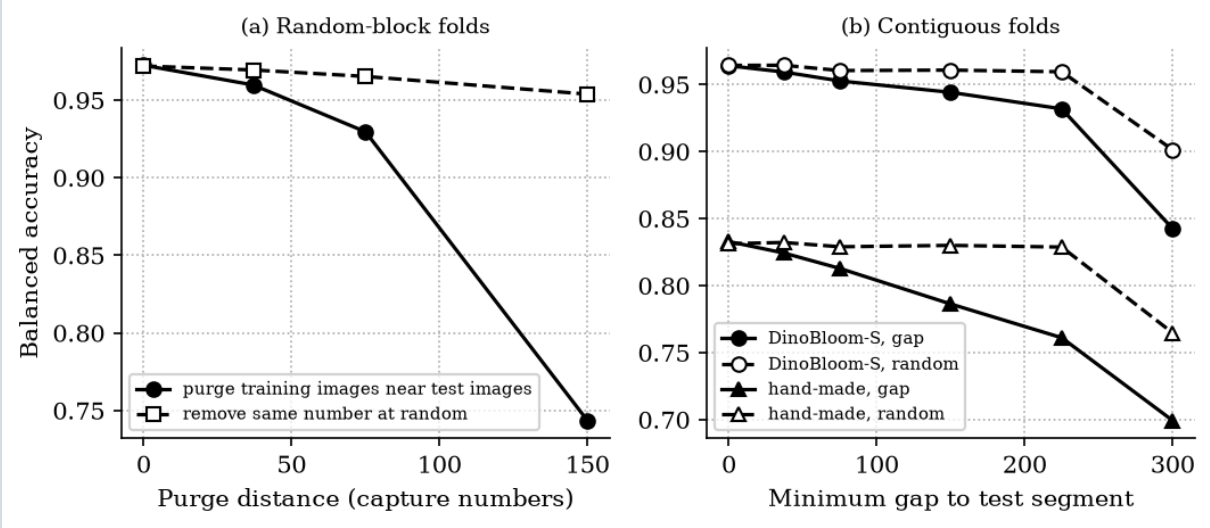


Figure R15. Accuracy against proximity in capture number. (a) Random-block folds: removing training images near the test images (solid) against removing the same number at random (dashed). (b) Contiguous folds: widening the gap to the test segment (solid) against random removal (dashed).

What it shows. If neighbouring images leak information, then deleting training images *near* the test images should hurt more than deleting the same number of training images *at random*. The gap between the solid and dashed lines measures that proximity effect.

> **What we see.** With random-block folds, purging training images within 75 capture numbers of the test images gives 0.930, while removing the same number at random gives 0.965; at a distance of 150 it is 0.744 against 0.954. With contiguous folds, a gap of 225 gives 0.932 (random removal 0.960) for the encoder features and 0.761 (random 0.829) for the hand-made features. At a gap of 300 even random removal falls (0.901) because the training sets become small.
>
> **What it means.** Nearby images do carry class-relevant information, so a random split overestimates accuracy, and the hand-made features depend on proximity more than the encoder does. The contiguous protocol with an embargo limits this effect but cannot remove it. These are single runs without confidence intervals, so they show a consistent pattern, not precise estimates.

R16. The GPU experiment: does fine-tuning the encoder help?

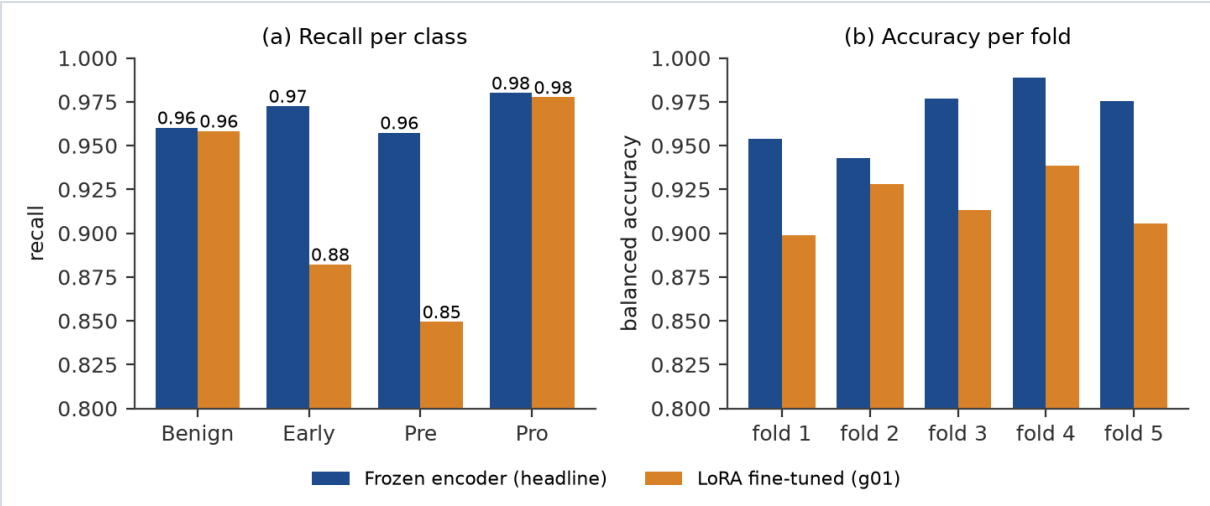


Figure R16a. Frozen encoder (headline) against low-rank adaptation (LoRA, run g01) trained on a GPU. (a) Recall per class. (b) Balanced accuracy per fold.

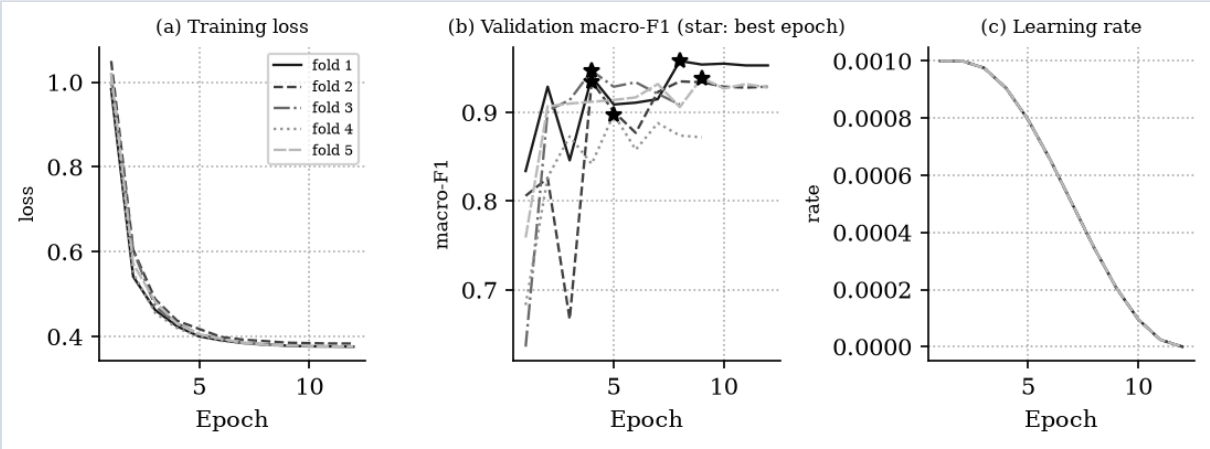


Figure R16b. Training curves of the LoRA run in the five folds: loss, validation macro-F1 (star = best epoch) and learning rate.

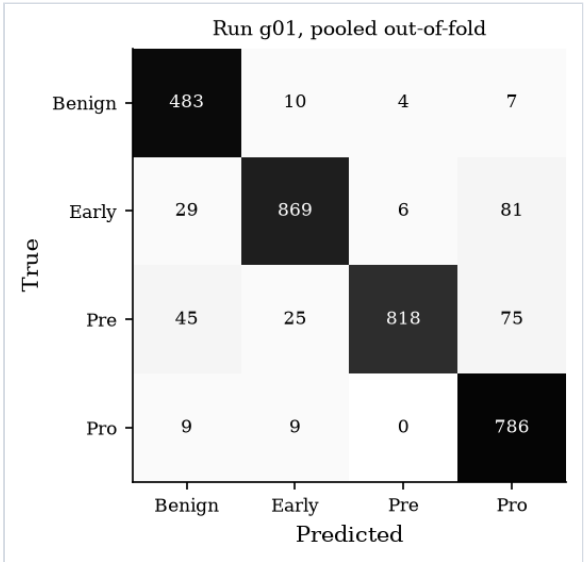


Figure R16c. Pooled out-of-fold confusion matrix of the LoRA run.

Model	Balanced accuracy	95% CI	Macro-F1	AUROC
Frozen DinoBloom-S + attention head (headline)	0.968	0.955 to 0.978	0.967	0.995
LoRA fine-tuned DinoBloom-S (g01, Kaggle T4 GPU)	0.917	0.898 to 0.935	0.907	0.988

What it shows. Low-rank adaptation (LoRA) lets the encoder itself learn a little (rank-8 updates in its attention layers) instead of staying frozen. The same five folds were used. Panel (a) compares recall per class; (b) compares each fold; R16b shows the LoRA training; R16c shows where the LoRA model makes mistakes.

What we see. The LoRA model is lower in **every** fold (for example fold 1: 0.899 against 0.954; fold 4: 0.939 against 0.989) and 5.1 points lower pooled. Benign (0.958) and Pro (0.978) are almost unchanged, but Early drops from 0.973 to 0.882 and Pre from 0.957 to 0.849. The confusion matrix shows the extra errors are mostly Early and Pre called Pro (81 and 75 images) and Pre called Benign (45). Each fold trained for 8 to 12 epochs, with the best epoch between 4 and 9, and the loss curves look healthy.

What it means. Adapting the encoder did *not* help here; the frozen features were not the limiting factor. The training curves show normal learning, so the drop is not a failed run. Two explanations are possible and neither was tested: the frozen encoder already holds what is needed, or the adapters picked up session-specific cues that do not hold across the embargo boundary. Only one of eight planned fine-tuning configurations was run to completion, so this result cannot say whether the drop is due to LoRA in particular, the encoder, or the settings.

R17. What the system actually outputs for an image

The table shows the real output of the fold-0 model for four images that this model never saw during training (one per class), with temperature-calibrated probabilities.

Image	True class	Cells found	Predicted	P(Benign)	P(Early)	P(Pre)	P(Pro)	Top cell attention
Benign-050	Benign	15	Benign	1.0000	0.0000	0.0000	0.0000	0.169
Early-031	Early	9	Early	0.0001	0.9997	0.0000	0.0001	0.177
Pre-170	Pre	22	Pre	0.0023	0.0004	0.9967	0.0006	0.098
Pro-031	Pro	5	Pro	0.0007	0.0001	0.0001	0.9991	0.780

What it shows. One row per image: how many cells were found, the answer, the probability of each class, and the largest attention weight.

What we see. All four are classified correctly with probabilities above 0.99. The Pro image has only 5 cells and one cell carries 78% of the attention; the Pre image has 22 cells and the attention is spread (largest weight 0.098).

What it means. This illustrates the output the GUI can present. Four hand-picked images are an illustration and not a performance estimate: performance is in the 3,256-image results above. High calibrated confidence on a dataset image does not carry over to images from other laboratories.

Results at a glance

Question	Answer	Where
How accurate is the headline model?	0.968 balanced accuracy	Block 6, R8, R9
Does it find malignant cases?	sensitivity 0.990, specificity 0.960	R8
Are the percentages honest?	ECE 0.010 after calibration	R10
Could it cheat with the background?	background alone: 0.823	R3, R14
Do blocks 2 to 4 carry information?	0.875 with simple classifier	R6
Does fine-tuning help?	no: 0.917 against 0.968	R16
Does evaluation design matter?	yes: up to 0.21 lower when near neighbours are removed	R15
What is the slowest block?	BEEMD, 83% of 904 ms	R5

Glossary

Term	Plain meaning
ALL	Acute lymphoblastic leukemia, a fast-growing cancer of immature lymphocytes.
Hematogone	A normal immature lymphocyte precursor that can resemble a leukemia cell; the Benign class.
HDS	Hybrid Diffusion-Steered filter: smooths noise but not edges, and yields an edge map.
APC	Adaptive Principal Curvature: a map of how sharply brightness bends, built from the two principal curvatures.
Watershed	A method that separates touching blobs by "flooding" from their centres.
EMD / BEEMD	Splitting an image into layers from fine to coarse; the bidimensional ensemble version repeats it with small noise and averages.
IMF	Intrinsic Mode Function: one layer produced by EMD. A "pure" IMF is one judged to hold real structure.
FBPTAT	The proposed recurrent network trained with fractional-order back-propagation (see block 5).
Encoder / DinoBloom-S	A pre-trained network that converts a cell picture into 384 numbers describing it.

Attention (MIL)	Learned weights saying how much each cell of an image contributes to the image-level decision.
Sensitivity / specificity	Share of malignant images correctly found / share of benign images correctly cleared.
Balanced accuracy	The average of the per-class hit rates.
Calibration (ECE)	Whether "90% confident" really means right nine times in ten; ECE is the average gap.
Embargo	Removing training images numerically adjacent to test images so that near-duplicates cannot leak across.